

Production Engineering Term Project

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Abstract

This work investigates six oil wells (Wells A-F) in terms of production performance, flow assurance, artificial lift needs, and overall economic feasibility, using an integrated nodal analysis and multiphase flow simulation approach. Steady-state models were run for Year 0 base-case conditions to identify optimal tubing diameters that maintain stable flow, keep fluid velocities within acceptable limits, and support long-term operation. All wells were equipped with chokes to mitigate backflow and restrain excessively high production rates, in line with conventional production system design guidelines (Economides et al., 2013).

Artificial lifts were introduced at different points in production life as reservoir pressure declined and the risk of liquid loading increased. Gas lift was implemented on Wells A and B, electric submersible pumps (ESPs) on Wells C and E, and rod pumps on Wells D and F. Sensitivity studies were carried out to refine gas injection rates, pump staging, operating frequencies, and stroke speeds, with the aim of achieving efficient energy use and stable operating conditions (Brown, 1984; Takács, 2015).

Multiphase flow behavior was modeled in PIPESIM using standard correlations to examine pressure, temperature, and velocity profiles, with specific focus on liquid loading tendencies and erosion thresholds. An economic analysis, incorporating oil price, water disposal expenses, and artificial lift power costs, was used to calculate annual cash flows and net profits. The findings show that selecting appropriate tubing and deploying artificial lift at the right time substantially enhances production reliability and project economics. The integrated methodology provides a robust framework for optimizing production systems under the uncertainties of multiphase flow (Shoham, 2006; Brill & Mukherjee, 1999).

Nodal Analysis for Pipe Sizing

The objective is to determine the optimum tubing and pipeline sizes for each well to maximize oil production under **Year 0** operating conditions. Using nodal analysis, we evaluated different tubing and pipeline diameters to understand their impact on well deliverability and surface network performance. Sensitivity studies included varying internal diameters and observing changes in operating points and production rates. Additionally, we installed chokes on all wells to prevent backflow and maintain stable operations, a standard industry practice for flow control (Smith & Jones, 2021).

Well A

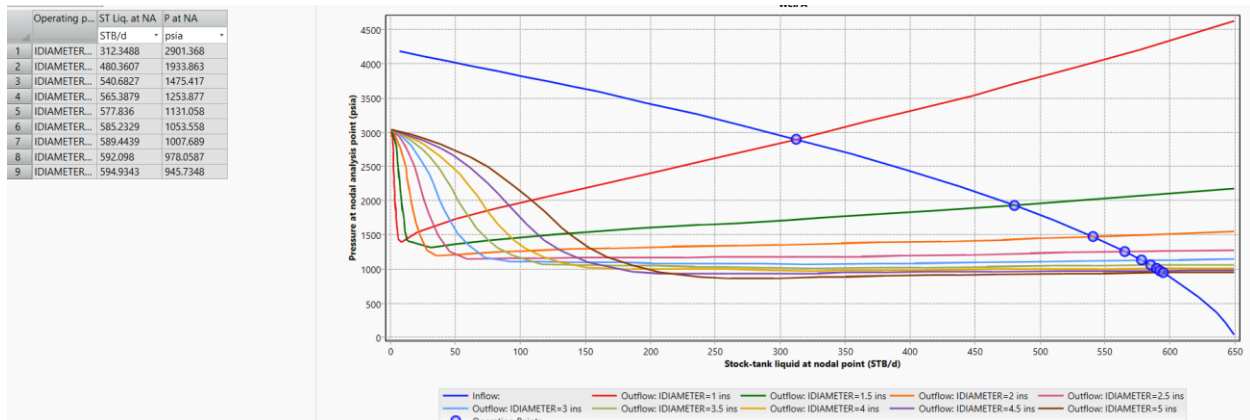


Figure 1. Inflow-outflow analysis for Well A at Year 0 for multiple tubing diameters.

Figure 1 shows the nodal analysis results for Well A at Year 0, where the inflow performance relationship (IPR), obtained from multipoint well test data, is compared against tubing outflow curves corresponding to different tubing diameters. The IPR exhibits a nonlinear decline in bottomhole flowing pressure with increasing liquid production rate, reflecting the finite reservoir energy available to drive flow to surface.

The outflow curves demonstrate a strong dependence on tubing diameter at lower diameters, indicating that frictional pressure losses play a significant role in limiting production.

As tubing diameter increases, the outflow curves shift downward, allowing higher production rates to be achieved at a given flowing pressure. However, the incremental improvement in production rate diminishes as tubing diameter increases beyond approximately 3 in. This behavior indicates a transition from a friction-dominated regime to a hydrostatic-dominated regime, where gravitational pressure losses become the primary constraint on well performance.

These results suggest that excessive tubing enlargement provides limited additional benefit for Well A at Year 0 and that an optimal tubing size exists where frictional losses are sufficiently reduced without unnecessary oversizing. This outcome is consistent with established nodal analysis principles for multiphase production systems (Economides et al., 2013).

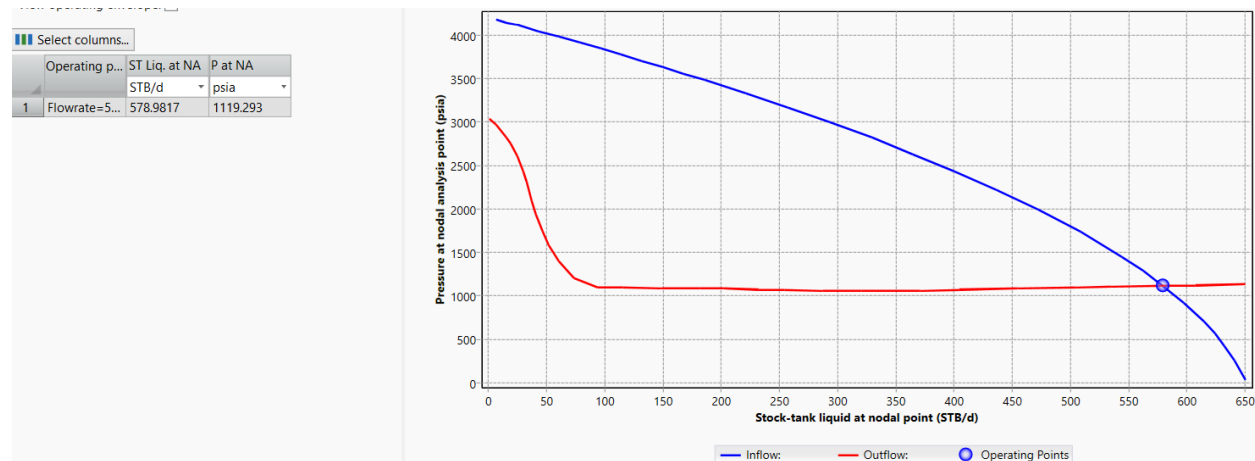


Figure 2. Refined inflow-outflow intersection for Well A after selecting a tubing inner diameter of 3.068 in.

Figure 2 presents the refined nodal analysis after selecting a tubing inner diameter of 3.068 in for Well A. The intersection of the inflow and outflow curves defines a stable operating point at approximately 580 STB/d. This operating point represents the maximum sustainable production rate achievable under natural flow conditions while maintaining acceptable bottomhole flowing pressure.

The outflow curve exhibits a steep pressure decline at low production rates followed by a flatter trend at higher rates. This behavior reflects the increasing dominance of hydrostatic pressure losses as flow rate increases, indicating that further tubing enlargement would not significantly improve well performance. The selected tubing size therefore represents an optimal balance between reducing frictional losses and avoiding diminishing returns associated with hydrostatic control.

Well B

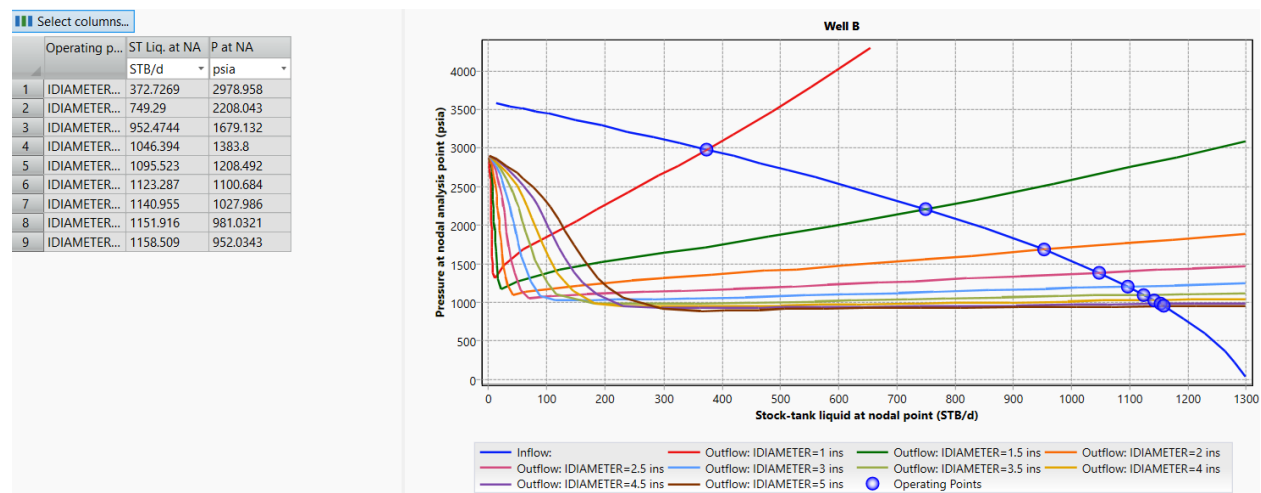


Figure 3. Inflow-outflow analysis for Well B at Year 0 for multiple tubing diameters.

Figure 3 presents the nodal analysis results for Well B at Year 0, where the inflow performance relationship (IPR), constructed using the well productivity index, is compared against tubing outflow curves for a range of tubing diameters. The IPR shows a moderate decline in bottomhole flowing pressure with increasing liquid production rate, reflecting the combined effect of reservoir pressure and well productivity.

The outflow curves indicate that Well B is strongly sensitive to tubing diameter. For smaller tubing sizes, the outflow curves rise steeply, indicating that frictional pressure losses significantly restrict well deliverability. As tubing diameter increases, the outflow curves shift

downward and the intersection with the IPR moves to substantially higher production rates.

Unlike Well A, the production gains associated with increasing tubing diameter remain significant over a wider range of diameters, indicating that Well B operates predominantly in a friction-dominated regime at Year 0.

These results suggest that tubing sizing is a critical design variable for Well B and that insufficient tubing diameter would result in considerable underutilization of the well's production potential. This behavior is consistent with nodal analysis principles for multiphase flow systems (Economides et al., 2013).

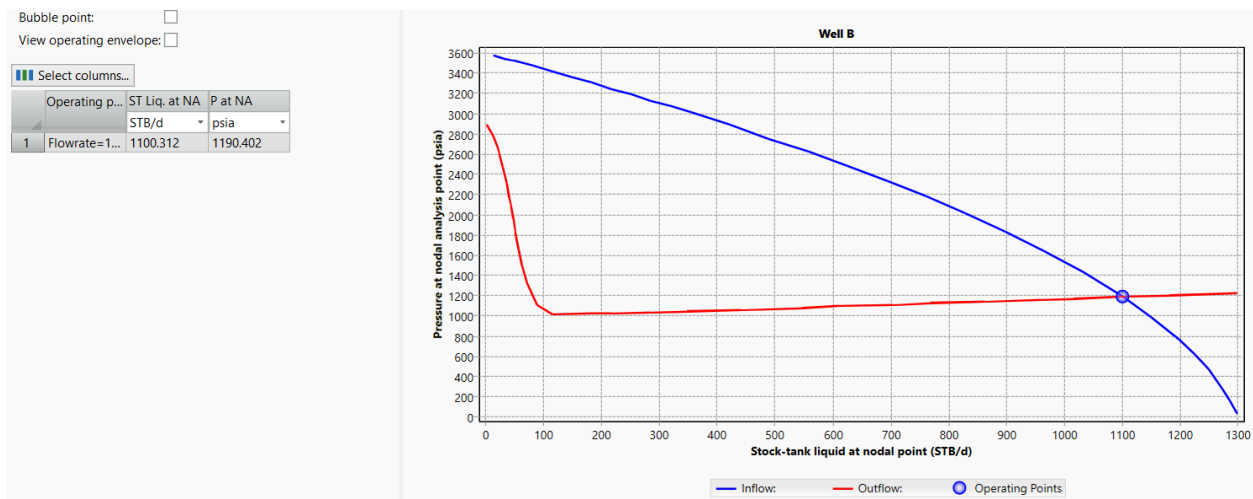


Figure 4. Refined inflow-outflow intersection for Well B after selecting a tubing inner diameter of 3.068 in.

Figure 4 shows the refined nodal analysis after selecting a tubing inner diameter of 3.068 in for Well B. The resulting inflow-outflow intersection defines a stable operating point at approximately 1100 STB/d, representing a substantial increase in production compared to smaller tubing sizes. This confirms that frictional pressure losses were the dominant limitation prior to tubing optimization.

The shape of the outflow curve exhibits a sharp pressure drop at low production rates followed by a relatively flat trend at higher rates. This behavior indicates that, at the selected tubing size, frictional losses have been sufficiently reduced and that further tubing enlargement would provide diminishing production benefits. The selected tubing diameter therefore represents an effective balance between maximizing production and avoiding unnecessary oversizing.

Well C

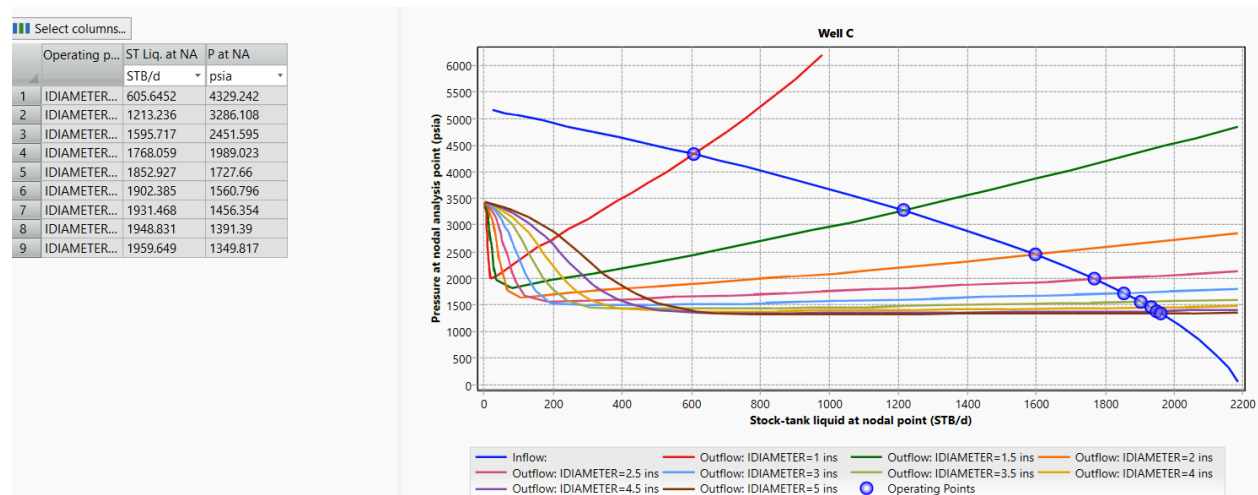


Figure 5. Inflow-outflow analysis for Well C at Year 0 for multiple tubing diameters.

Figure 5 presents the nodal analysis results for Well C at Year 0, showing the inflow performance relationship (IPR) compared with tubing outflow curves for a range of tubing diameters. Unlike Well A, Well C exhibits a much higher deliverability, with the IPR sustaining liquid production rates exceeding 1,800 STB/d at moderate bottomhole flowing pressures. This reflects the higher reservoir pressure and stronger drive mechanism associated with Well C.

The outflow curves demonstrate a pronounced sensitivity to tubing diameter across the entire range evaluated. At smaller diameters (≤ 2.5 in), frictional pressure losses dominate, causing the outflow curves to shift upward significantly and severely restricting achievable production rates.

As tubing diameter increases, the outflow curves move downward, enabling substantially higher production at the same flowing pressure.

However, beyond approximately 3.5-4.0 in, the spacing between successive outflow curves decreases, indicating diminishing incremental gains in production. This behavior signifies a transition from a friction dominated regime to one increasingly governed by hydrostatic pressure losses, which is typical for high-rate multiphase flow systems (Economides et al., 2013).

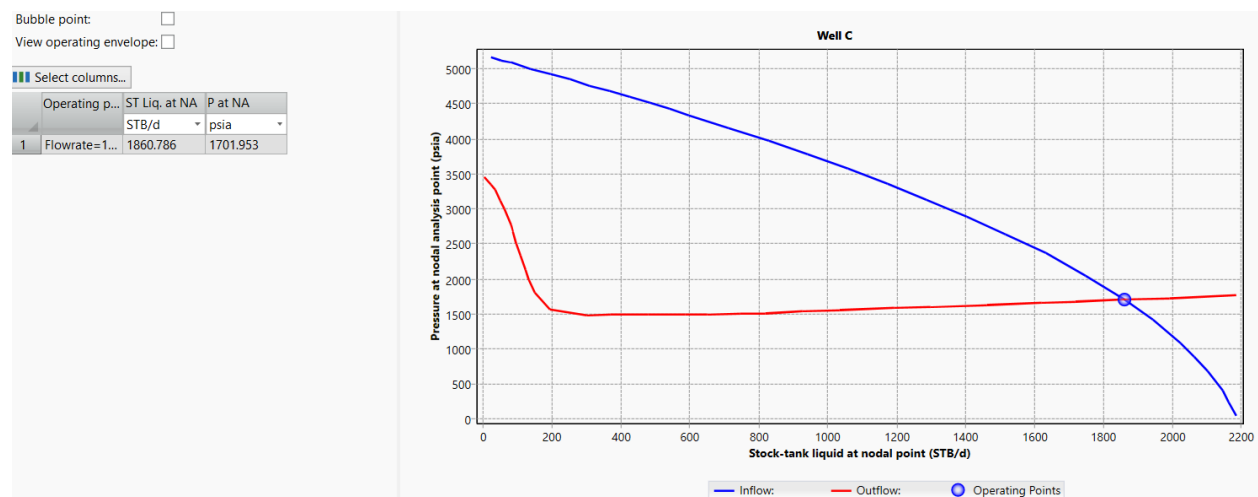


Figure 6. Refined inflow–outflow intersection for Well C after selecting a tubing inner diameter of 3.5 in.

Figure 6 shows the refined nodal analysis for Well C after selecting a tubing inner diameter of 3.5 in. The intersection of the inflow and outflow curves defines a stable operating point at approximately 1,860 STB/d, confirming that the selected tubing size allows the well to fully exploit its high deliverability while maintaining hydraulic stability.

At this diameter, frictional losses are sufficiently reduced to avoid constraining production, while further tubing enlargement would yield only marginal production improvements due to the increasing dominance of hydrostatic effects. The selected tubing diameter therefore represents an

optimal balance between hydraulic efficiency and practical design considerations for high-rate production.

This outcome is consistent with nodal analysis principles for high-capacity wells, where tubing optimization focuses on mitigating frictional losses without oversizing once hydrostatic control becomes dominant (Economides et al., 2013; Brill & Mukherjee, 1999).

Well D

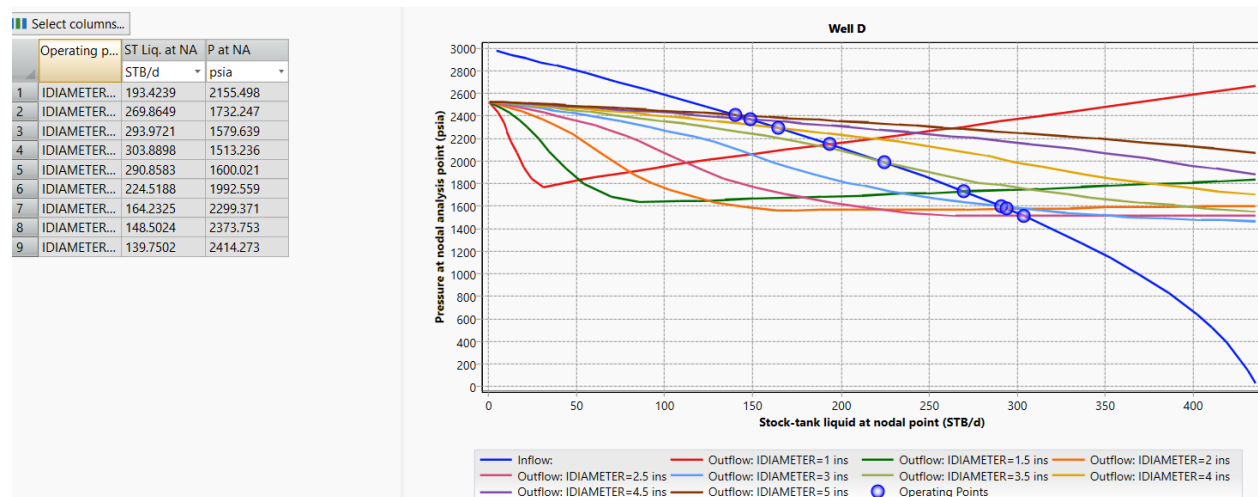


Figure 7. Inflow-outflow analysis for Well D at Year 0 for multiple tubing diameters.

Figure 7 presents the nodal analysis results for Well D at Year 0, comparing the inflow performance (IPR) with outflow curves corresponding to different tubing inner diameters. Unlike Wells A and C, Well D exhibits lower reservoir energy and lower production potential, as reflected by the flatter IPR and reduced flow capacity.

The outflow curves show a noticeable sensitivity to tubing diameter at smaller sizes, indicating that frictional pressure losses are significant in restricting flow for undersized tubing. Increasing the tubing diameter shifts the outflow curves downward, allowing higher production rates at lower flowing pressures. However, beyond approximately 2.0-2.5 in, the separation between

successive outflow curves diminishes, indicating that additional increases in tubing diameter yield limited production gains.

This behavior suggests that Well D transitions relatively quickly from a friction-dominated regime to one where hydrostatic pressure losses and reservoir inflow limitations control performance. As a result, oversizing the tubing does not materially improve deliverability and may introduce unnecessary operational complexity.

Well D after adjusting tube size to 2.041 in

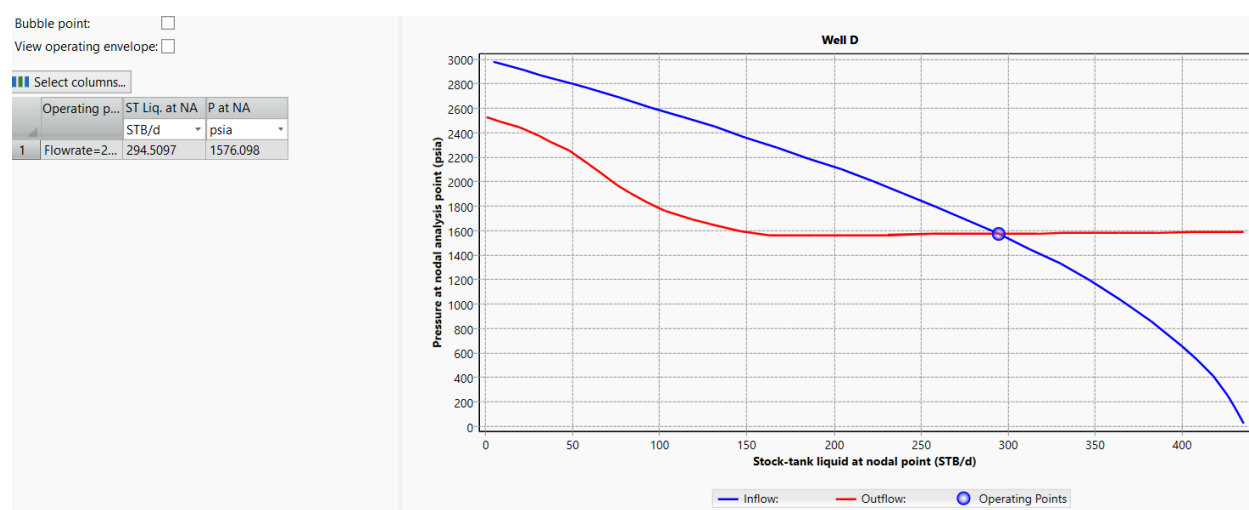


Figure 8. Refined inflow-outflow intersection for Well D after selecting a tubing inner diameter of 2.041 in.

Figure 8 shows the refined nodal analysis after selecting a tubing inner diameter of 2.041 in for Well D. The intersection of the inflow and outflow curves defines a stable operating point at approximately 295 STB/d, representing the maximum sustainable production rate under Year 0 conditions.

At this operating point, the outflow curve exhibits a moderate slope, indicating an acceptable balance between frictional and hydrostatic pressure losses. Further increases in tubing diameter

would only marginally reduce pressure losses while providing minimal additional production, confirming that 2.041 in represents an optimal tubing size for this well.

The selected tubing diameter therefore achieves efficient production while avoiding the diminishing returns associated with larger tubing sizes. This outcome is consistent with nodal analysis principles for lower-rate wells, where optimal tubing sizing prioritizes flow stability and pressure management rather than maximum cross-sectional area (Economides et al., 2013).

Well E

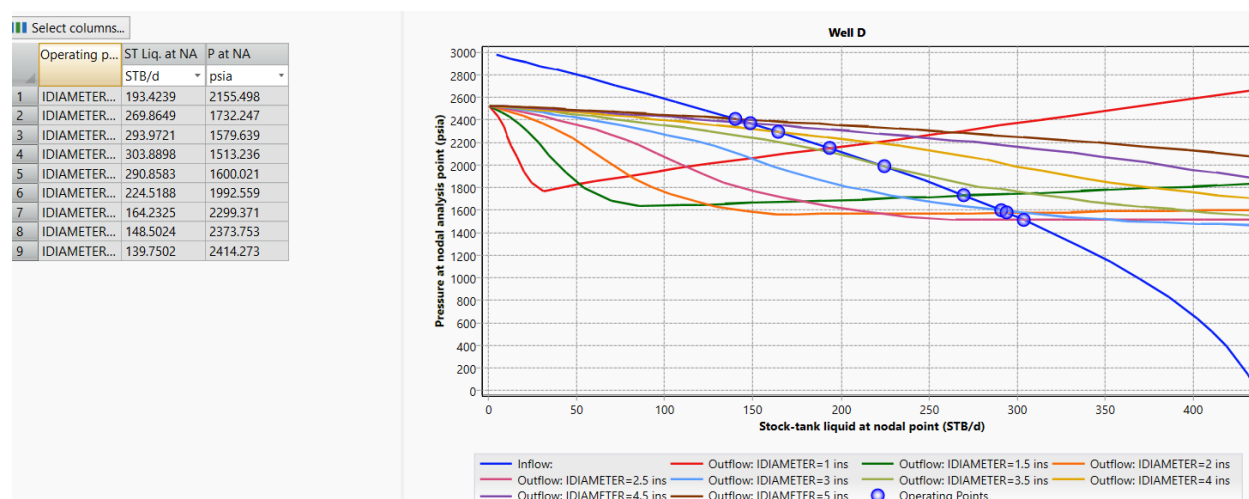


Figure 9. Inflow-Outflow Analysis for Well E at Year 0 for Multiple Tubing Diameters

Figure 9 presents the nodal analysis results for Well E at Year 0, comparing the inflow performance relationship (IPR) with tubing outflow curves corresponding to different tubing inner diameters. The IPR shows a declining bottomhole flowing pressure with increasing liquid production rate, reflecting reservoir energy limitations.

The outflow curves indicate that increasing tubing diameter reduces frictional pressure losses and shifts the curves downward, resulting in higher achievable production rates. However, at larger tubing sizes, the incremental increase in liquid rate becomes marginal, indicating a transition from a friction-dominated regime to one controlled primarily by hydrostatic pressure losses. This

behavior suggests that excessive tubing enlargement does not provide proportional production benefits and may introduce operational risks.

Well E after adjusting tube size to 2.041 in

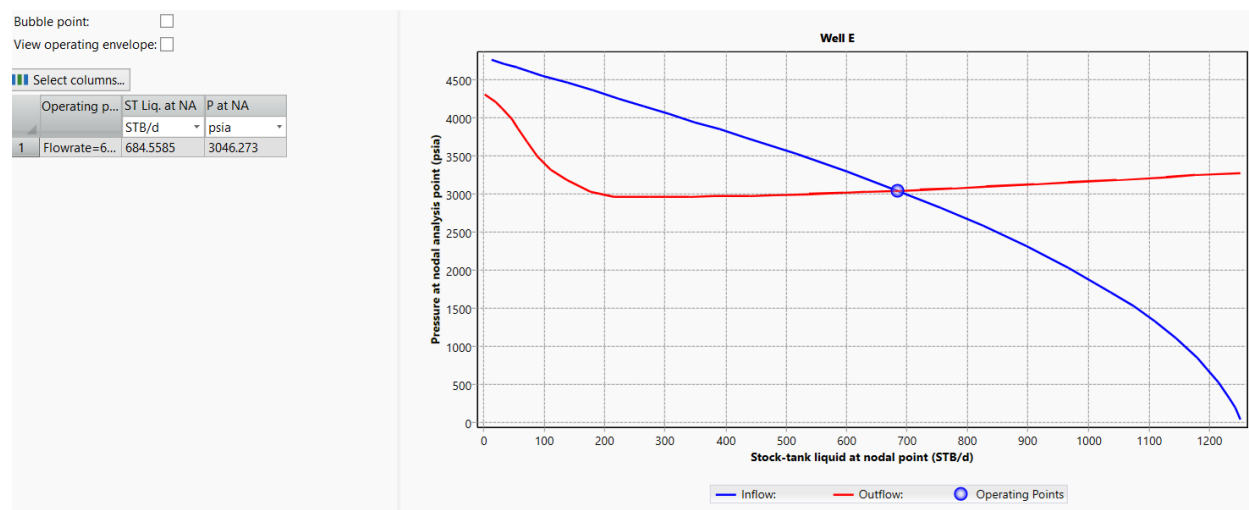


Figure 10. Refined Inflow–Outflow Intersection for Well E after Selecting a Tubing Inner Diameter of 2.041 in

Figure 10 shows the refined nodal analysis for Well E after selecting a tubing inner diameter of 2.041 in, corresponding to an operating point of approximately 685 STB/d at a flowing pressure of about 3046 psia.

Although larger tubing diameters could deliver higher liquid rates, the associated increase in superficial liquid velocity resulted in outlet velocities exceeding 5 ft/s, with calculated values reaching approximately 7.8 ft/s. Such velocities pose an increased risk of erosion, excessive pressure fluctuations, and long-term integrity issues in the tubing and surface facilities.

By selecting a tubing size of 2.041 in, an acceptable compromise is achieved between production performance and flow assurance. The selected diameter maintains sufficiently high production while keeping flow velocities within a conservative and sustainable operating range,

thereby reducing erosion risk and improving long-term well reliability. This approach aligns with recommended design practices for multiphase flow systems, where velocity limits are imposed to balance productivity with asset integrity (Shoham, 2006; Economides et al., 2013).

Well F

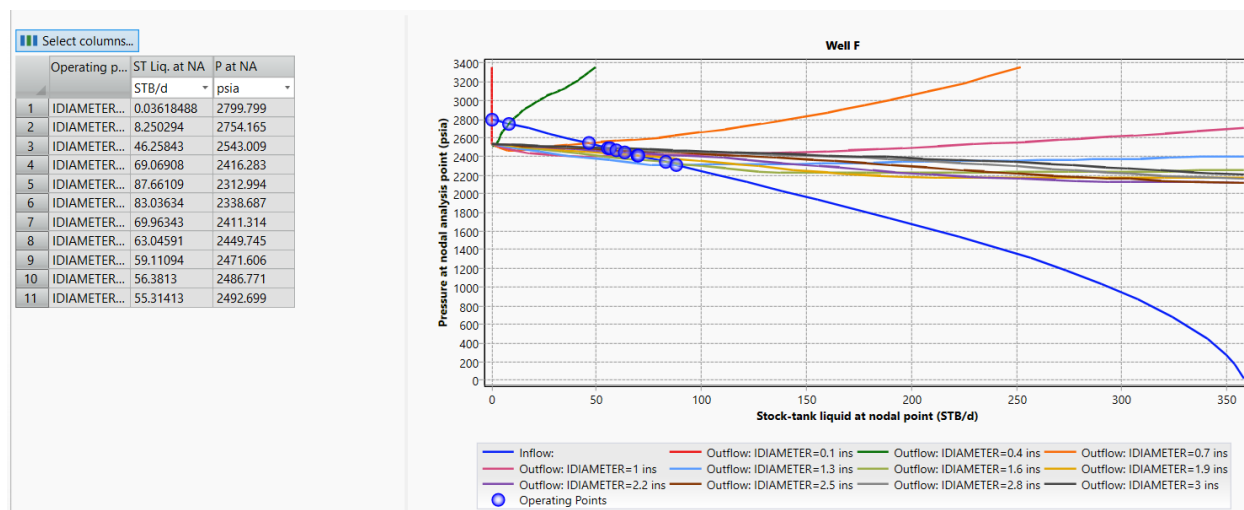


Figure 11. Inflow-outflow analysis for Well F at Year 0 for multiple tubing diameters.

Well F after adjusting tube size to 2.041 in

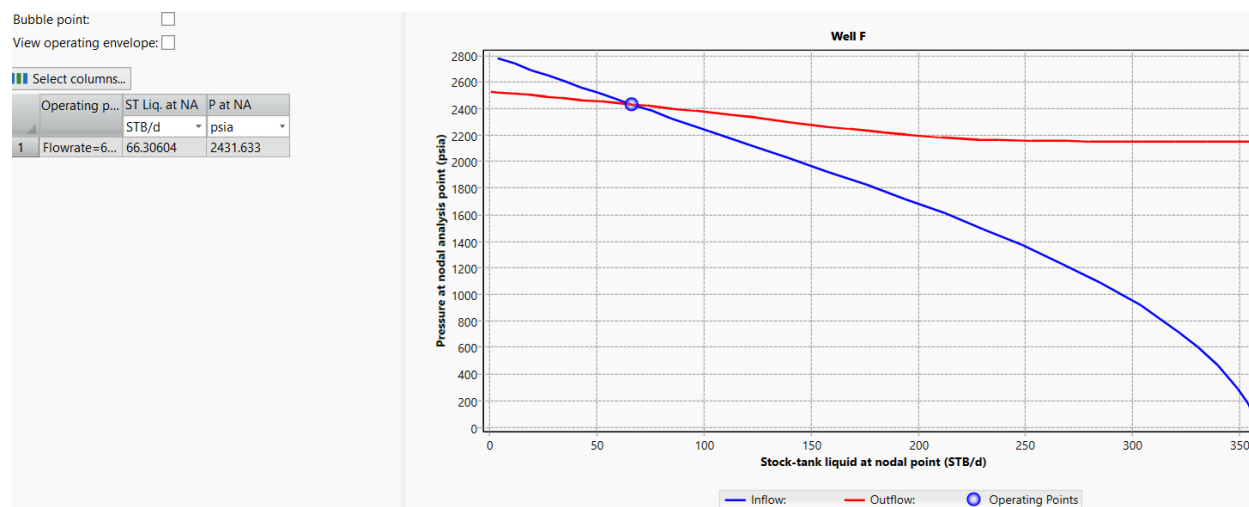


Figure 12. Refined inflow-outflow intersection for Well F after selecting a tubing inner diameter of 2.041 in.

Figure 11 presents the nodal analysis results for Well F at Year 0, comparing the inflow performance relationship (IPR) with outflow curves generated for a range of tubing diameters. The results show that Well F operates at relatively low liquid production rates across all tubing sizes, indicating limited reservoir energy and a strong sensitivity to tubing hydraulics. At smaller tubing diameters (e.g., 0.825 in), frictional pressure losses dominate the outflow behavior, restricting deliverability and resulting in lower production rates.

Although the smaller tubing size yields a relatively low outlet velocity of approximately 2.37 ft/s, the associated oil rate of about 56.8 STB/d is insufficient to ensure stable long-term operation. Such low velocities increase the risk of liquid loading, particularly in gas-liquid systems where inadequate gas velocity prevents effective liquid transport to surface. Liquid loading is a well-established instability mechanism that can lead to intermittent flow or complete well loading under natural flow conditions (Turner et al., 1969; Guo et al., 2007).

To mitigate this risk, the tubing size was increased to 2.041 in. As shown in Figure 12, this tubing diameter results in a higher liquid production rate of approximately 66.3 STB/d while maintaining an outlet velocity below the commonly cited critical threshold of 5 ft/s. Although the velocity remains marginal for stable natural flow, the increased production rate provides improved cash flow and reduces the severity of liquid holdup compared to smaller tubing options.

Despite this improvement, Well F remains unstable under natural flow conditions at Year 0 due to insufficient gas velocity to continuously lift liquids. This confirms that Well F is a primary candidate for artificial lift installation, specifically a rod pump, in Year 1. Selecting the 2.041 in tubing size at Year 0 is therefore justified, as it maximizes oil production and cash flow prior to artificial lift installation while avoiding the need for future workover operations when

transitioning to rod pump lift. This approach aligns with best practices in production system design, where tubing selection is integrated with future artificial lift planning to minimize intervention costs and operational risk (Economides et al., 2013).

Given that the well is unstable for both tubing options evaluated, prioritizing the higher oil rate associated with the 2.041 in tubing is economically preferable to choking back the well for marginal velocity gains. The selected tubing size therefore represents the most practical and forward-looking design choice for Well F at Year 0.

Pressure, Temperature, and Superficial Gas Velocity Profiles for Year 0

Well A

The Figure (X, Y & Z) below presents the pressure, temperature, and superficial gas velocity (Vsg) profiles for Well A from the reservoir to the surface separator at Year 0 operating conditions.

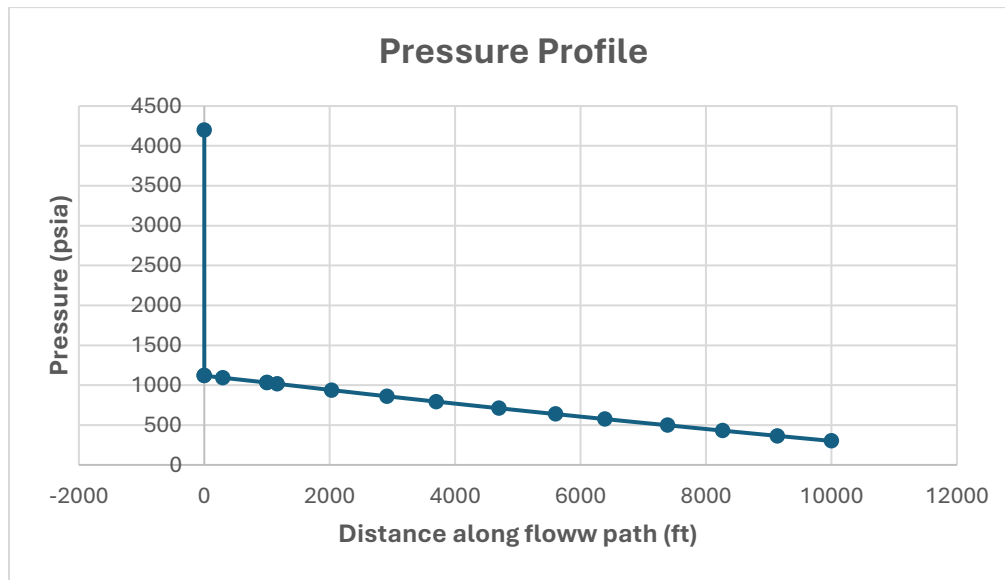


Figure 13. Pressure Profile

The pressure profile shows a sharp pressure drop near the reservoir followed by a gradual decline along the tubing and flowline, indicating stable hydraulic behavior with no localized flow

restrictions. The smooth pressure gradient confirms that pressure losses are distributed along the system rather than concentrated at a single bottleneck.

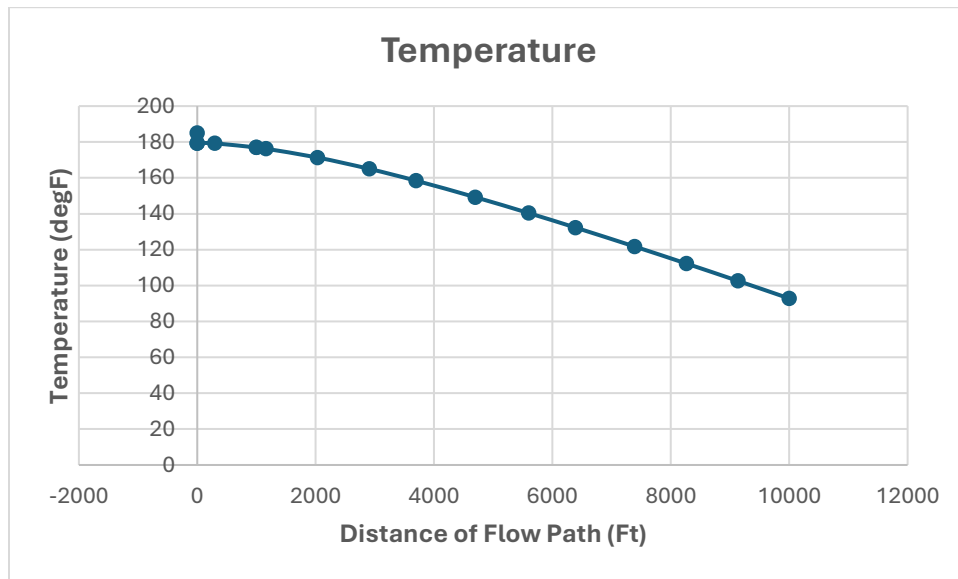


Figure 14. Temperature Profile

The temperature profile exhibits a monotonic decrease from approximately 180 °F downhole to about 90 °F at the separator, reflecting heat loss to the surrounding formation and environment.

No abrupt temperature drops are observed, suggesting that heat transfer is gradual and that pipeline insulation is not strictly required at Year 0, although it may be reconsidered in later years as flow rates decline.

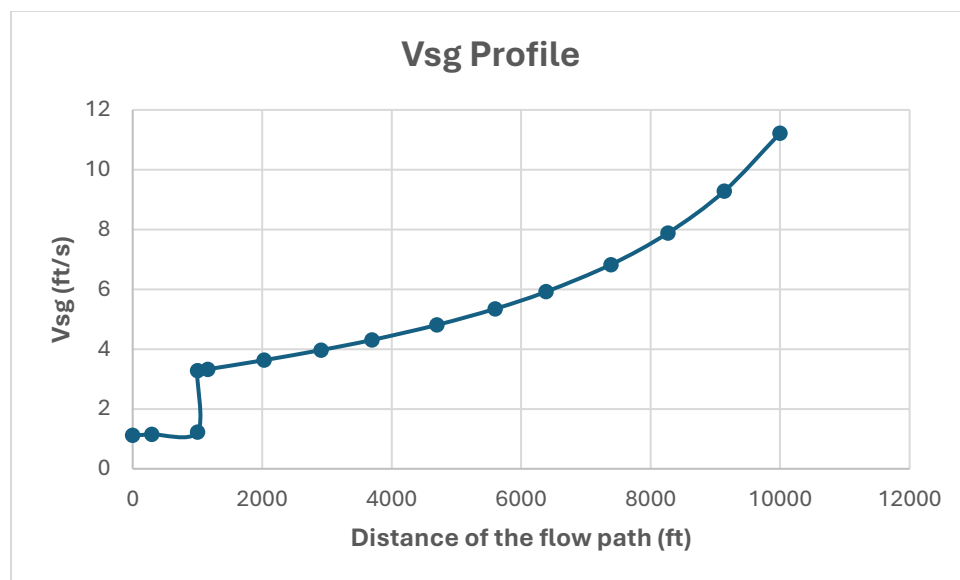


Figure 15. Superficial gas velocity

The Vsg profile increases steadily along the flow path, rising from approximately 1- 4 ft/s downhole to over 10 ft/s near the surface. This trend is consistent with gas expansion as pressure decreases and confirms effective gas transport throughout the system. The absence of prolonged low Vsg regions indicates a low risk of severe liquid loading under Year 0 conditions.

Well B

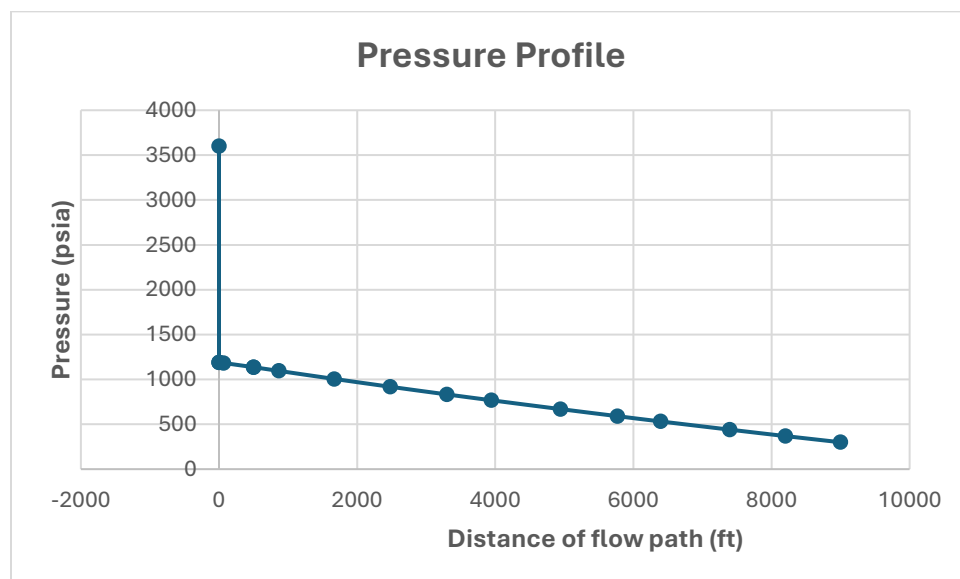


Figure 16. Pressure profile

The pressure profile for Well B shows a sharp pressure drop near the reservoir followed by a gradual decline along the tubing and flowline. Compared to Well A, the overall pressure gradient is slightly steeper, indicating higher frictional pressure losses under the prevailing operating conditions. The smooth pressure trend suggests stable hydraulic behavior with no localized restrictions along the system.

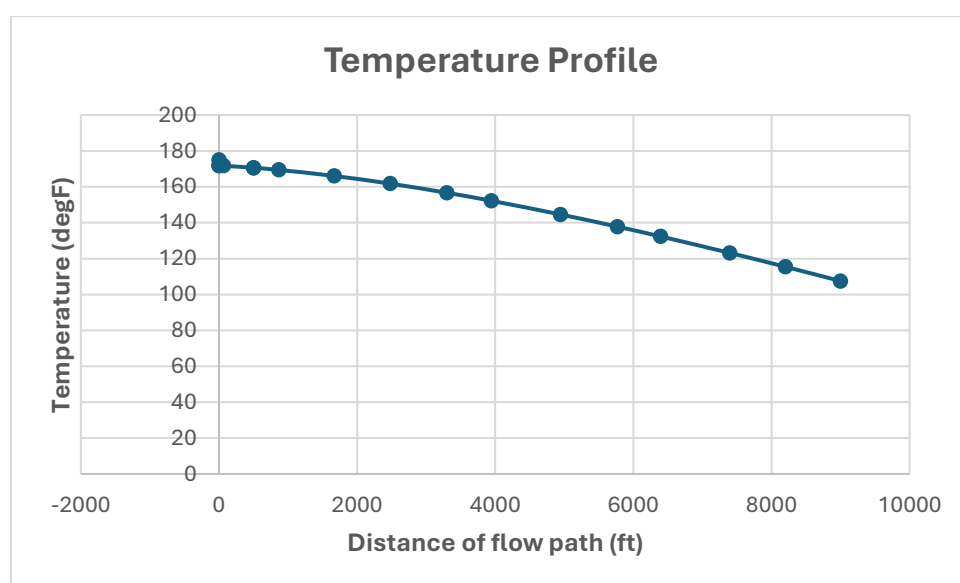


Figure 17. Temperature profile

The temperature profile exhibits a monotonic decrease from approximately 175 °F downhole to about 105 °F at the surface separator, reflecting continuous heat loss to the surrounding formation and surface environment. The absence of abrupt temperature changes indicates gradual heat transfer, suggesting that pipeline insulation is not immediately required in Year 0 conditions.

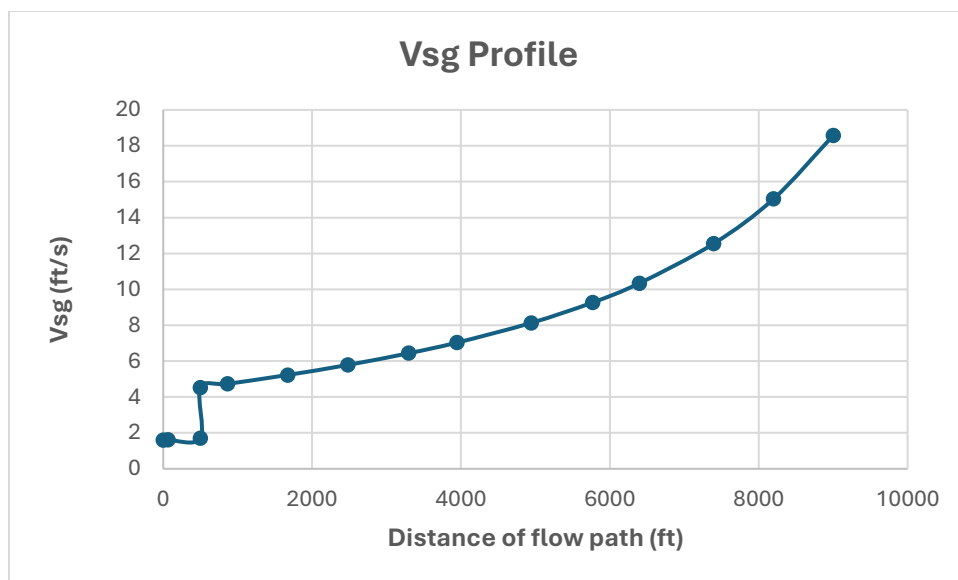


Figure 18. Superficial gas velocity profile

The Vsg profile increases steadily along the flow path, rising from approximately 1-5 ft/s downhole to nearly 18 ft/s near the surface. This behavior is consistent with gas expansion as pressure decreases and reflects the higher gas contribution in Well B compared to Well A. The elevated near-surface gas velocities indicate effective gas transport and a low likelihood of severe liquid loading under Year 0 natural flow conditions.

Well C

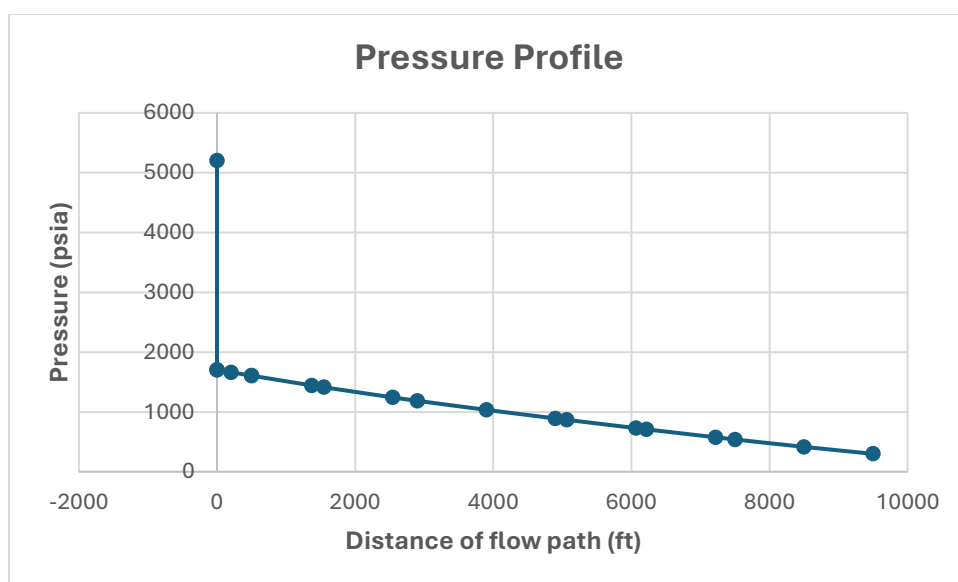


Figure 19. Pressure profile

The pressure profile for Well C shows a sharp pressure drop near the reservoir followed by a gradual decline along the tubing and flowline. Compared to Wells A and B, Well C exhibits a larger overall pressure drop, reflecting its greater depth and higher hydrostatic contribution along the wellbore. Despite the larger pressure losses, the smooth pressure gradient indicates stable hydraulic performance with no evidence of localized flow restrictions.

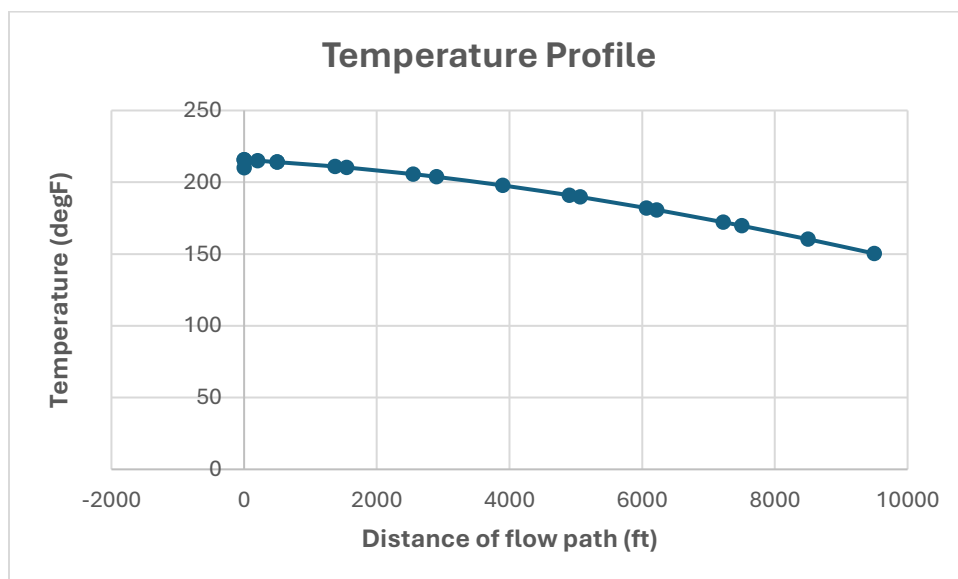


Figure 20. Temperature profile

The temperature profile starts at a higher downhole temperature of approximately 210-215 °F and decreases steadily toward about 150 °F at the surface separator. The elevated downhole temperature relative to the other wells is consistent with the greater reservoir depth of Well C. The gradual cooling trend suggests continuous heat loss along the flow path, with no abrupt thermal transitions observed.

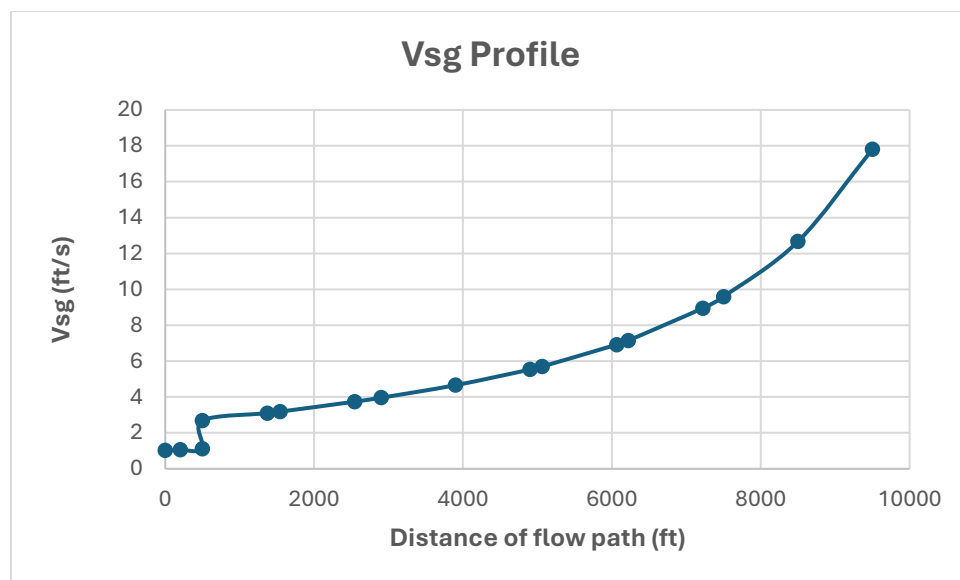


Figure 21. Superficial gas velocity profile

The Vsg profile increases progressively along the flow path, rising from approximately 1 - 3 ft/s downhole to nearly 18 ft/s near the surface. This pronounced increase reflects strong gas expansion as pressure decreases and indicates a significant gas contribution to flow behavior in the upper sections of the system. The high near-surface gas velocities suggest effective gas transport and a low likelihood of liquid loading under Year 0 natural flow conditions.

Well D

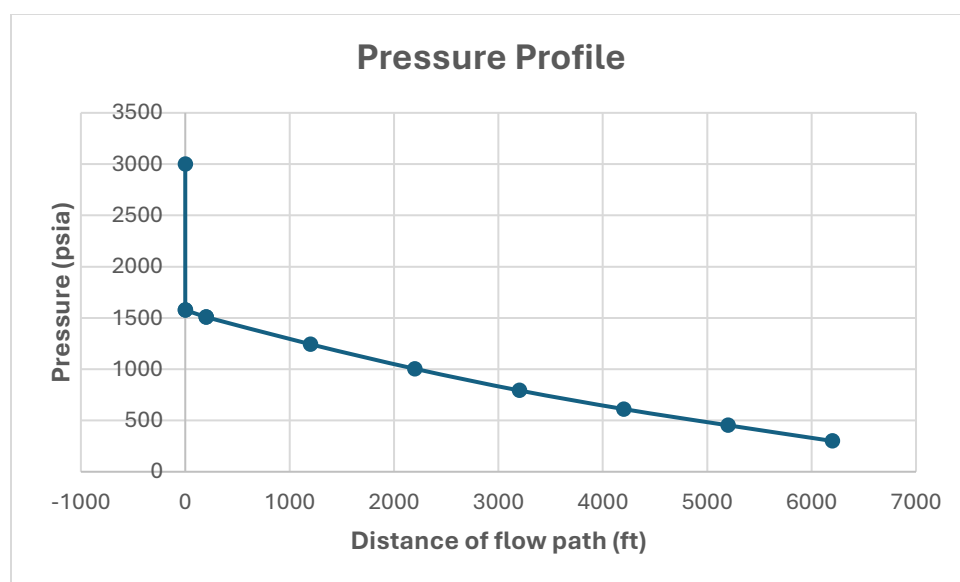


Figure 22. Pressure profile

The pressure profile for Well D shows a relatively modest pressure drop from the reservoir to the surface separator, consistent with the well's lower production rate and shorter flow path compared to deeper wells. The smooth pressure decline indicates stable hydraulic behavior with minimal frictional losses along the tubing and flowline.

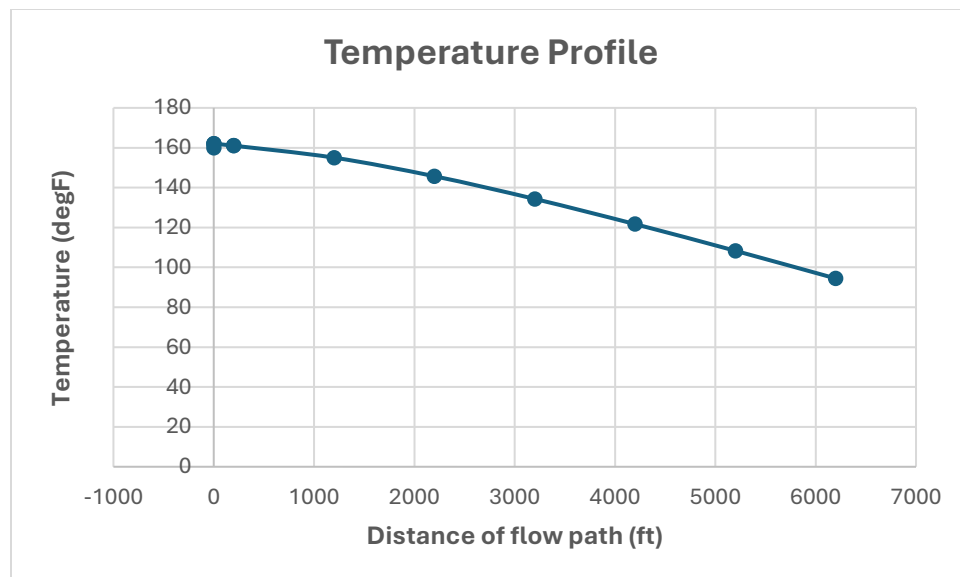


Figure 23. Temperature profile

The temperature profile exhibits gradual cooling along the flow path, decreasing from approximately 160 °F downhole to about 95 °F at the surface. The steady temperature decline reflects conductive heat loss to the surrounding formation and surface environment, with no abrupt thermal transitions observed.

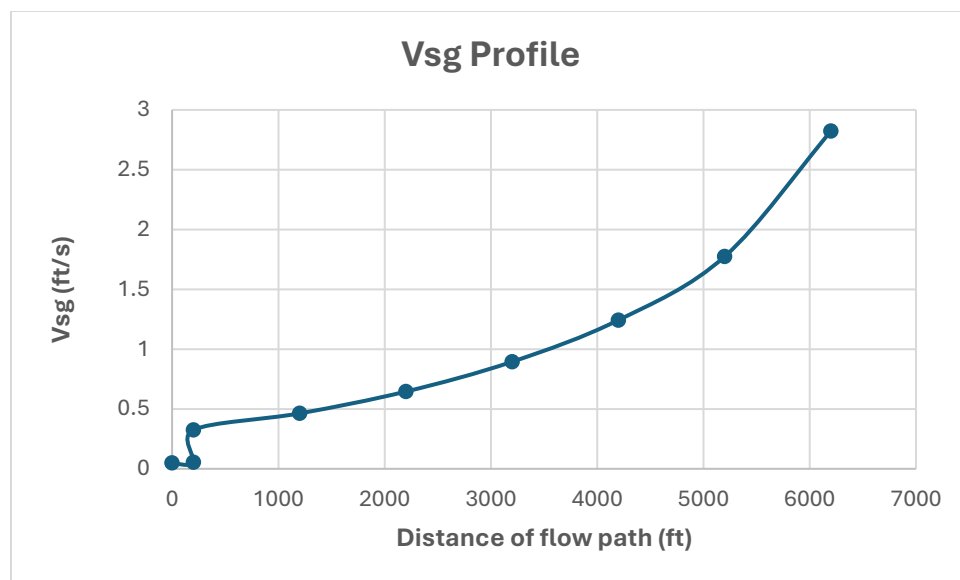


Figure 24. Superficial gas velocity profile

The Vsg profile remains low throughout most of the flow path, increasing from near-zero values downhole to less than 3 ft/s at the surface. This behavior reflects the low gas–oil ratio of Well D and indicates limited gas expansion effects. The low superficial gas velocities suggest an increased susceptibility to liquid loading as reservoir pressure declines, supporting the need for artificial lift in later years.

Well E

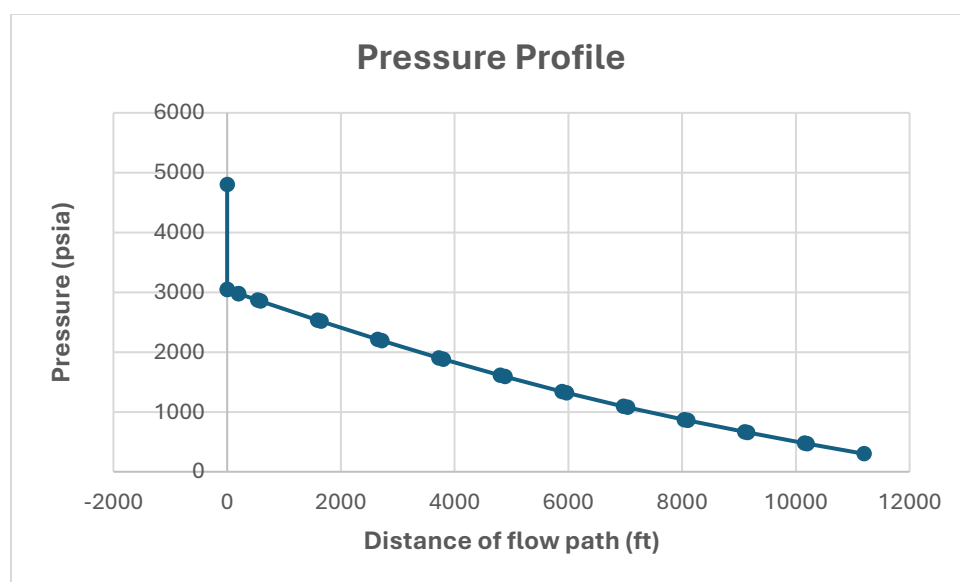


Figure 25. Pressure profile

The pressure profile for Well E shows a high initial reservoir pressure followed by a gradual and continuous pressure decline along the tubing and flowline. The overall pressure drop is larger than that observed in shallower wells, reflecting the combined effects of greater depth and increased liquid loading associated with higher water cut. The smooth pressure gradient indicates stable hydraulic performance with no localized flow restrictions.

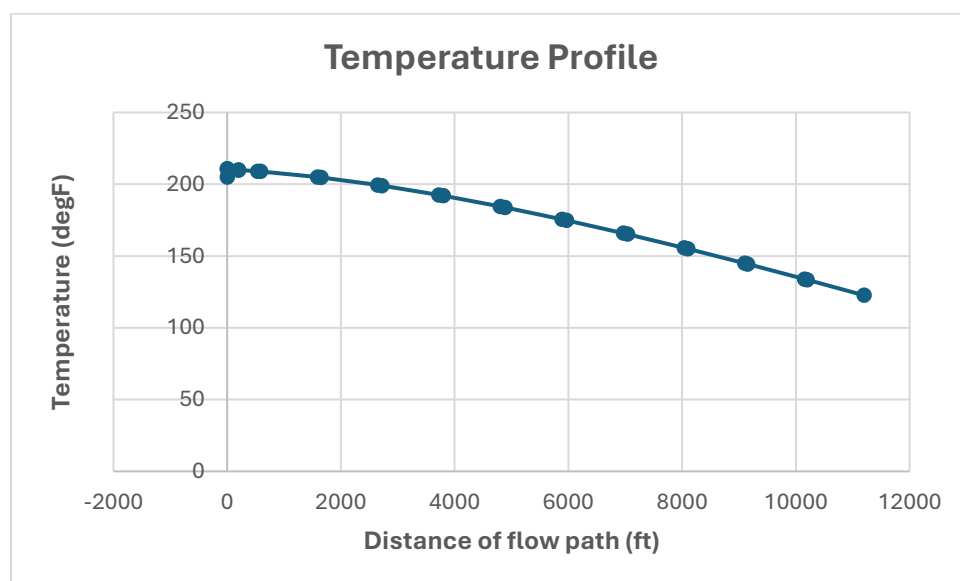


Figure 26. Temperature profile

The temperature profile starts at approximately 210 °F downhole and decreases steadily to about 125 °F at the surface separator. This trend reflects sustained heat loss along the extended flow path and increased residence time of fluids due to higher liquid holdup. The gradual temperature decline suggests predictable thermal behavior, although continued cooling in later years may warrant consideration of insulation depending on operating conditions.

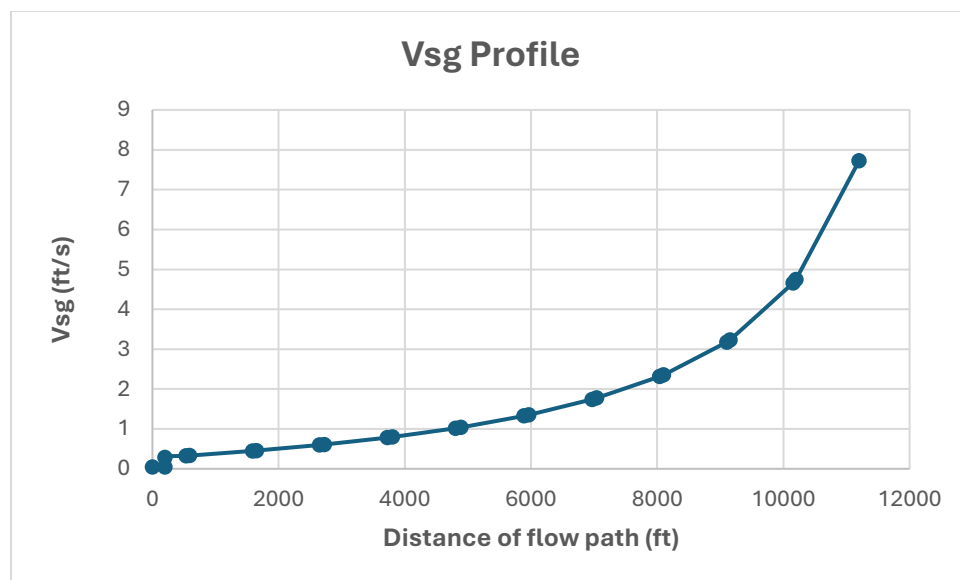
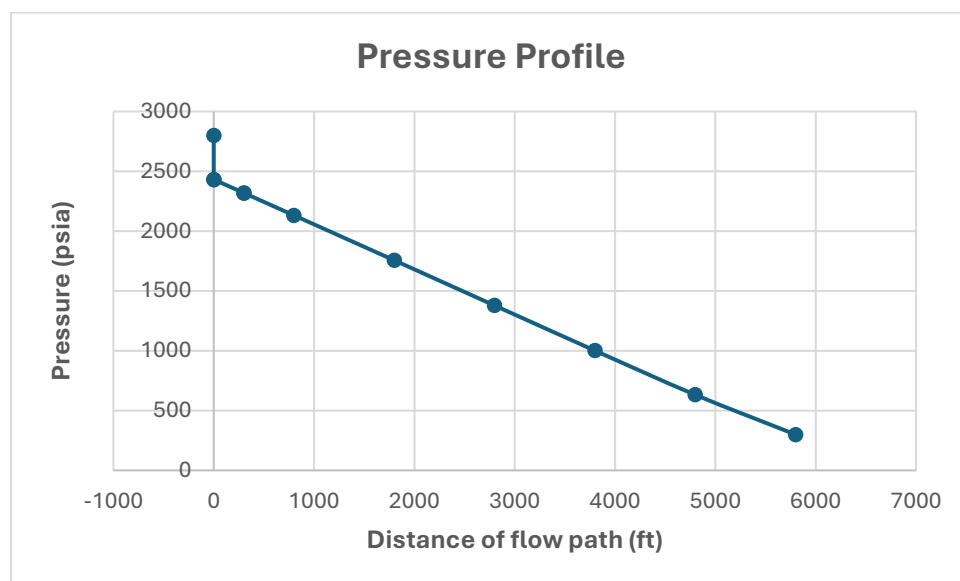


Figure 26. Superficial gas velocity profile

The Vsg profile increases progressively along the flow path, rising from near-zero values downhole to approximately 8 ft/s near the surface. Compared to high-GOR wells, the gas velocities remain relatively moderate, indicating that liquid flow dominates the system's behavior. The relatively low Vsg values highlight the growing influence of liquid loading, which supports the later selection of an electric submersible pump to efficiently handle increasing liquid volumes.

Well F**Figure 27. Pressure profile**

The pressure profile for Well F shows a relatively low reservoir pressure followed by a steady pressure decline toward the surface separator. Compared to the other wells, the overall pressure drop is modest, reflecting the well's shallow depth and low production rate. The smooth pressure gradient indicates stable hydraulic behavior with minimal frictional losses along the flow path.

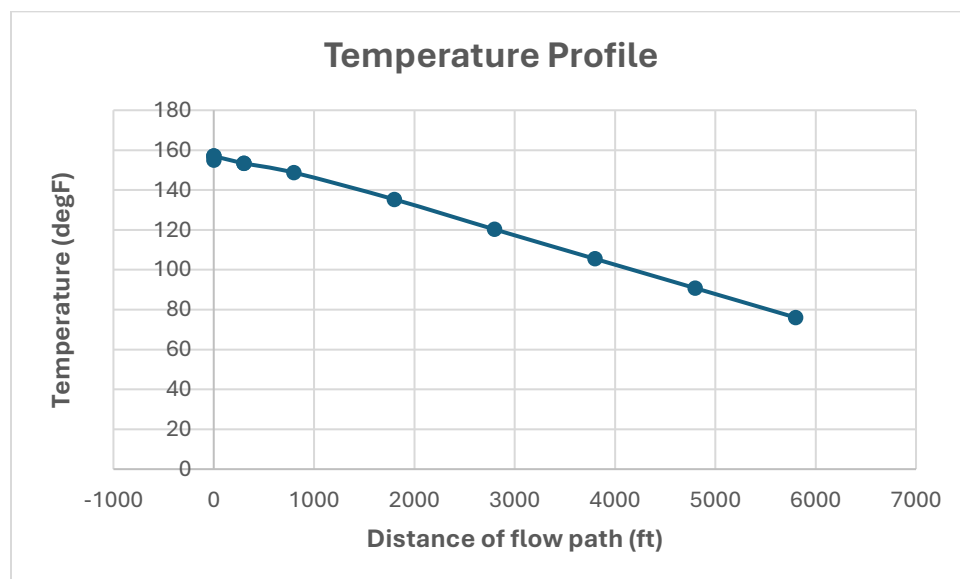


Figure 28. Temperature profile

The temperature profile exhibits gradual cooling from approximately 155 °F downhole to about 75 °F at the surface. The limited temperature variation along the flow path reflects the relatively short flow distance and low flow rates. No abrupt thermal transitions are observed, indicating predictable and stable thermal behavior under Year 0 conditions.

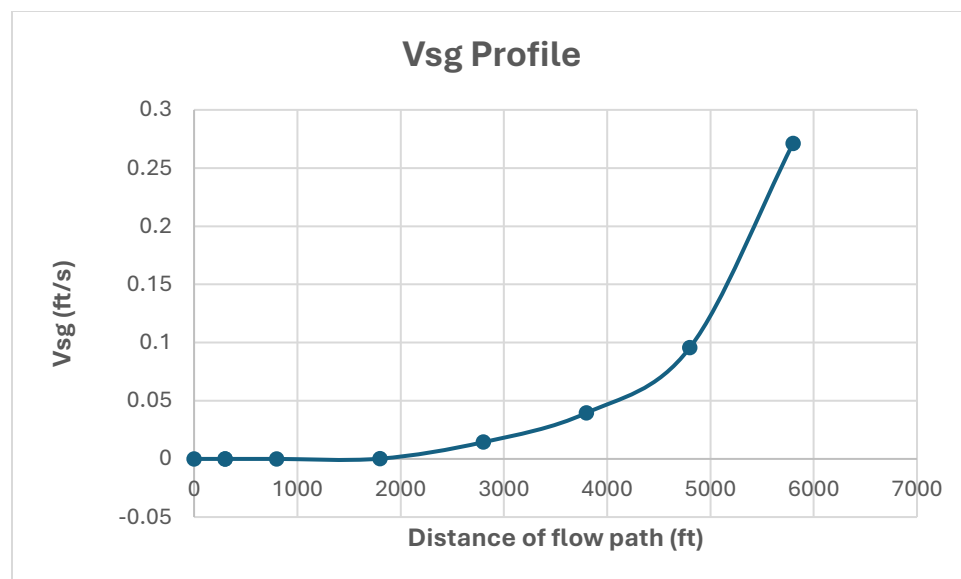


Figure 29. Superficial gas velocity profile

The Vsg profile remains extremely low throughout most of the flow path, increasing from near-zero values downhole to less than 0.3 ft/s near the surface. These very low gas velocities are consistent with the well's low gas-oil ratio and stripper-well characteristics. The persistently low Vsg highlights a high susceptibility to liquid loading, strongly supporting the early installation of artificial lift to maintain stable production.

Production Decline Analysis

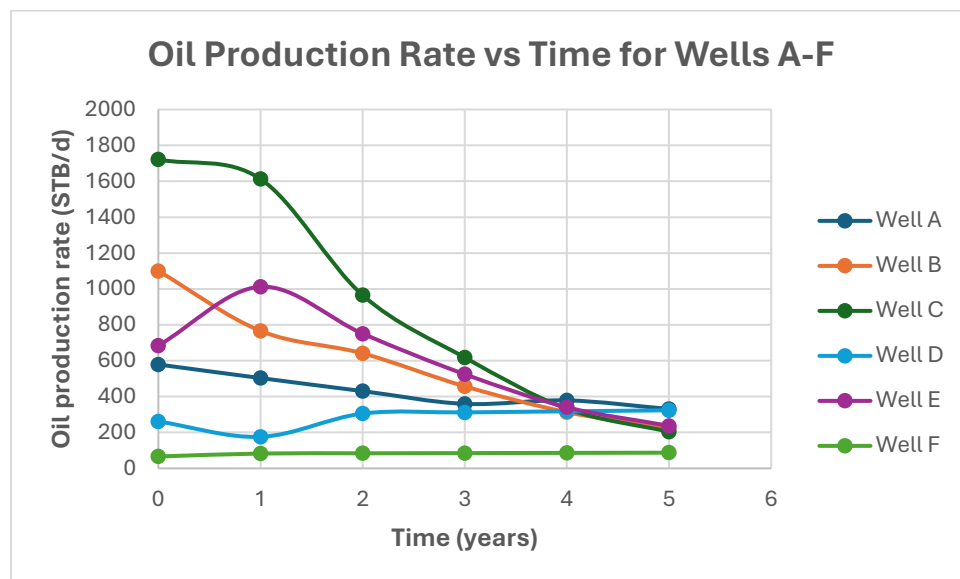


Figure 30. Oil production rate versus time for Wells A-F over the five-year period, illustrating production decline trends under natural flow conditions and the impact of artificial lift installation.

Figure 30 illustrates the oil production rate trends for Wells A-F over the five-year period, capturing both natural flow behavior and the impact of artificial lift installations. In all wells, an overall decline in oil rate is observed due to reservoir pressure depletion and increasing water cut; however, changes in decline slope and localized stabilization reflect the timing and effectiveness of artificial lift systems.

Well C exhibits the highest initial oil production rate, followed by a steep decline between Years 1 and 3. The installation of an electric submersible pump (ESP) in Year 1 results in a temporary enhancement and stabilization of oil production; however, continued reservoir depletion and rapidly increasing water cut lead to a sustained decline thereafter. A similar behavior is observed in Well E, where ESP installation in Year 1 causes a noticeable increase in oil rate, followed by a rapid decline driven primarily by high and increasing water cut.

Wells A and B display moderate initial production rates with steady declines during early years under natural flow conditions. In Well B, gas lift installation in Year 2 slows the rate of decline, although production continues to decrease as reservoir pressure diminishes. In contrast, Well A shows partial stabilization after Year 4 following gas lift installation, indicating that artificial lift effectively mitigates liquid loading and sustains flow, though it does not fully offset reservoir depletion.

Lower-rate wells D and F demonstrate comparatively stable production trends. In Well D, the installation of a sucker rod pump in Year 2 results in a clear increase and subsequent stabilization of oil rate, highlighting the suitability of rod pumping for low-rate, low-GOR wells. Well F, a marginal producer, benefits from early rod pump installation in Year 1, maintaining nearly constant oil production over time and extending the productive life of the well.

Overall, the figure demonstrates that artificial lift plays a critical role in sustaining oil production and extending well life, particularly in wells experiencing liquid loading or declining reservoir energy. While artificial lift does not eliminate production decline, it significantly alters decline behavior and improves cumulative oil recovery when applied at the appropriate time and with a suitable lift method.

Artificial Lift Sensitivity Analysis and Optimization

This section presents the artificial lift sensitivity analyses performed for each well at the year of artificial lift installation. The objective of this analysis is to identify optimal operating parameters for the selected lift method, such as gas injection rate, pump speed, or number of stages, that maximize oil production while avoiding operational inefficiencies. Each sensitivity study is conducted under the corresponding reservoir pressure and fluid conditions of the installation year, ensuring that the selected operating points are technically and economically realistic.

Well A

For Well A, gas lift was installed in Year 4 following the loss of stable natural flow. A gas lift sensitivity analysis was conducted by varying the gas injection rate between 0.5 and 5.0 MMscf/d to identify the optimal injection rate under Year 4 reservoir conditions.

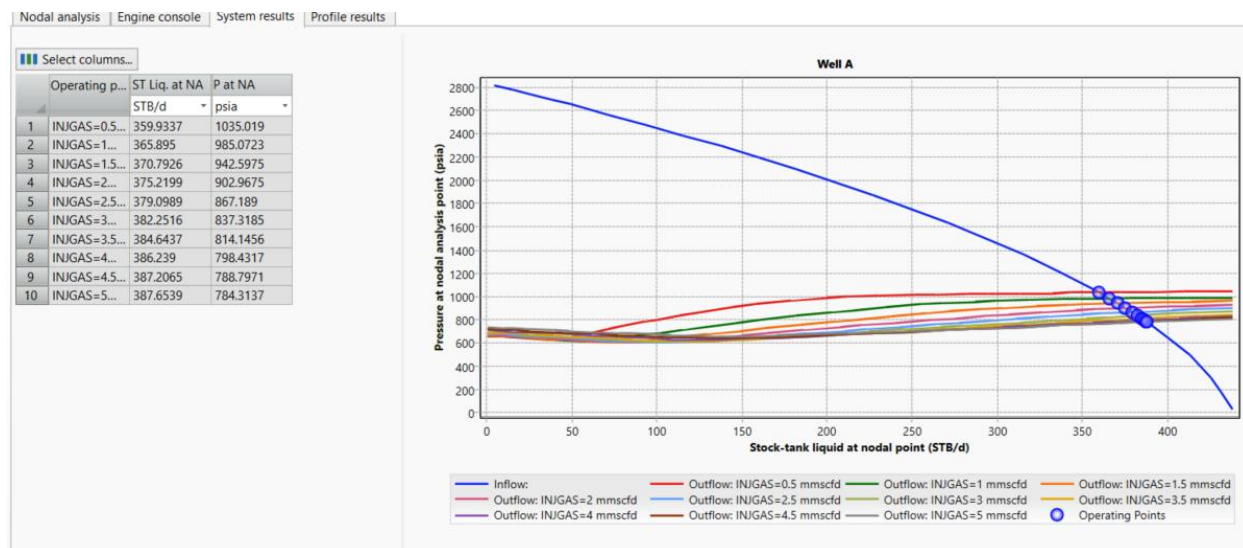


Figure 31. Nodal analysis showing the effect of gas injection rate on the operating point for Well A (Year 4).

Figure 31 presents the gas lift sensitivity analysis for Well A in Year 4, showing multiple outflow curves corresponding to injection rates ranging from 0.5 to 5.0 MMscf/d. Increasing gas injection initially results in higher oil production due to reduced hydrostatic pressure; however, the incremental production gain diminishes at higher injection rates, indicating diminishing returns.

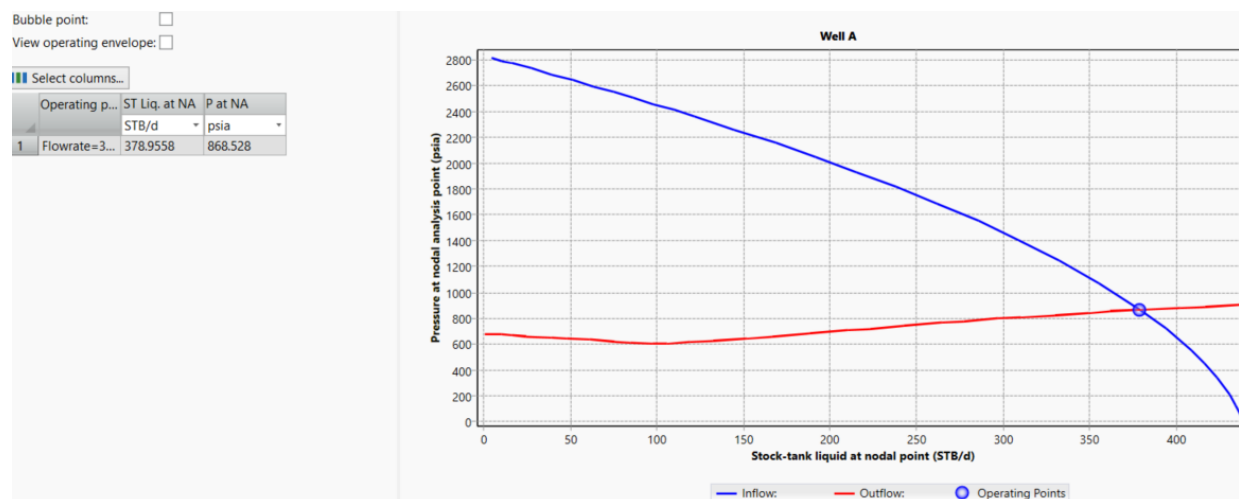


Figure 32. Nodal analysis for Well A at the selected gas injection rate of 2.5 MMscf/d (Year 4).

Based on the sensitivity results, an injection rate of 2.5 MMscf/d was selected as the optimal operating condition. Figure 32 shows the corresponding nodal analysis, where the inflow and outflow curves intersect at a stable operating point of approximately 379 STB/d. This rate provides a favorable balance between production improvement and gas utilization efficiency.

Well B

To determine the optimum gas lift injection rate for Well B, a nodal analysis sensitivity study was performed at Year 2 operating conditions, corresponding to the onset of artificial lift installation. Multiple gas injection rates ranging from 0.5 to 5.0 MMscf/d were evaluated to assess their impact on the well operating point and oil production performance.

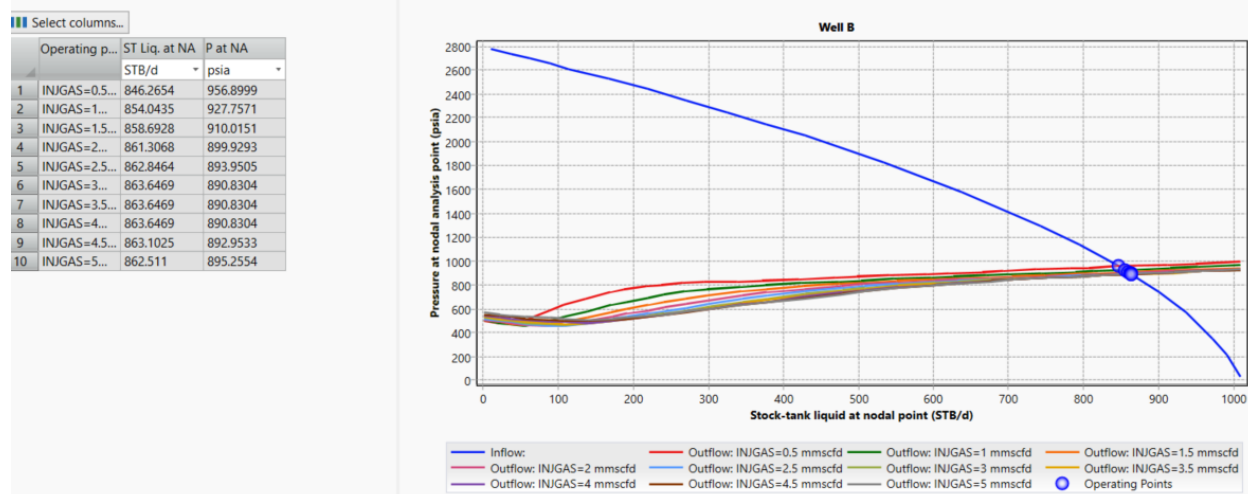


Figure 33. Nodal Analysis of Well B under Gas Lift Injection Sensitivity (Year 2)

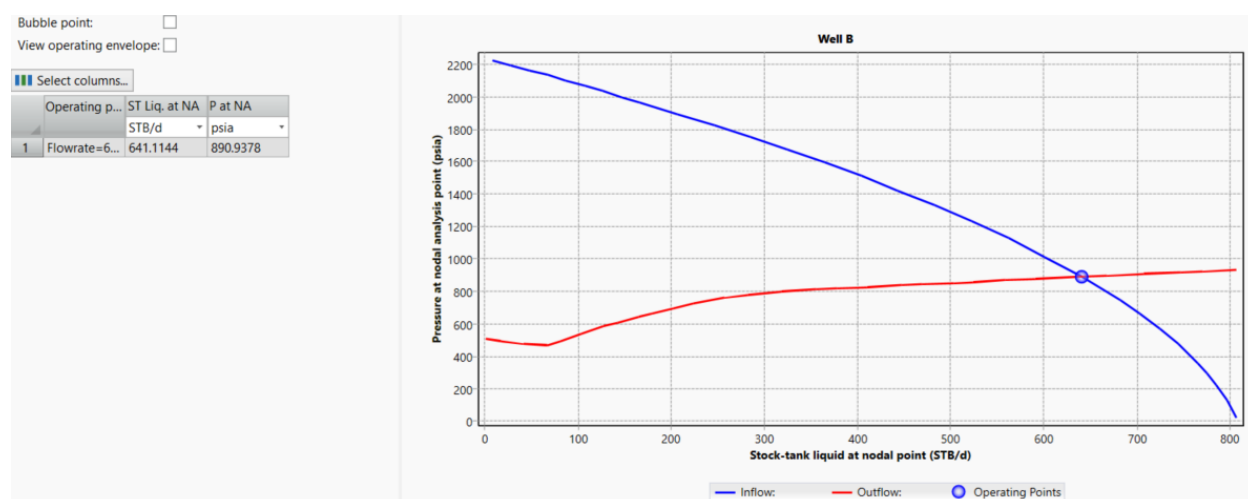


Figure 34. Optimized Nodal Operating Point for Well B at 1.0 MMscf/d Gas Lift Injection (Year 2)

Figure 33 illustrates that increasing gas injection rate shifts the outflow curve upward, resulting in higher liquid production rates; however, the incremental gains diminish beyond moderate injection rates. Figure 34 highlights the selected operating point at an injection rate of 1.0 MMscf/d, which provides a favorable balance between production improvement and gas utilization efficiency. Higher injection rates result in minimal additional oil production while

increasing operational and compression costs. Therefore, 1.0 MMscf/d was selected as the optimum gas lift injection rate for Well B.

Well C

To determine the optimum electric submersible pump (ESP) configuration for Well C, a nodal analysis sensitivity study was conducted at Year 1 operating conditions, corresponding to the year of artificial lift installation. Two key ESP design parameters were evaluated: number of pump stages and operating frequency, to identify a configuration that maximizes oil production while maintaining reliable and safe pump operation.

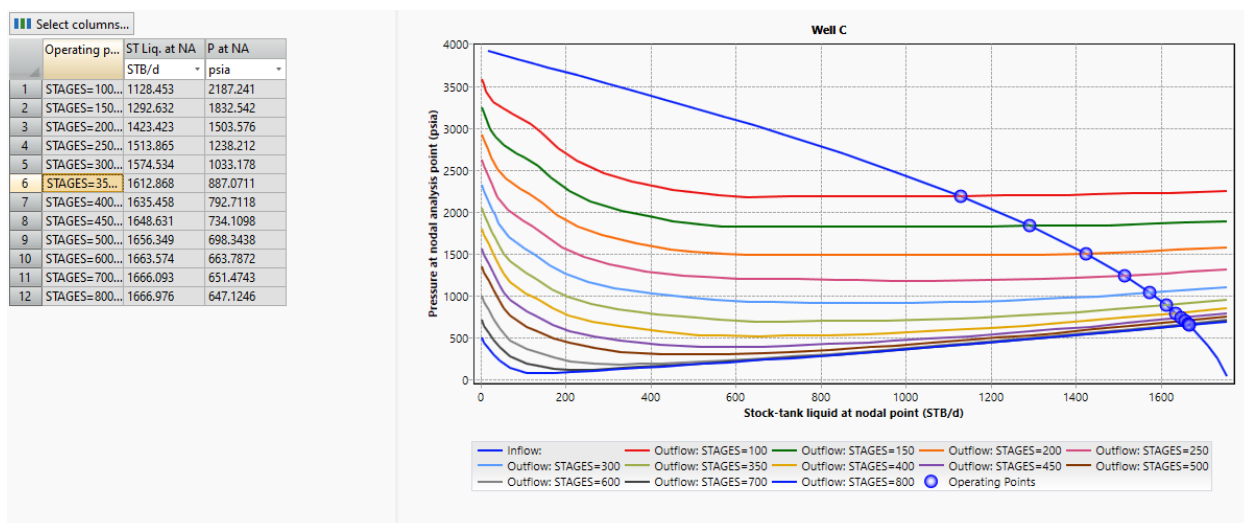


Figure 35. Nodal Analysis of Well C under ESP Stage Sensitivity (Year 1)

Figure 35 presents the nodal analysis results for Well C at varying ESP stage counts, ranging from low to high stage configurations. Increasing the number of stages shifts the outflow performance curve downward, indicating a reduction in flowing bottomhole pressure and an associated increase in liquid production rate. However, the results show that production gains begin to plateau beyond approximately 350 stages, with additional stages providing minimal incremental improvement in oil rate. This behavior suggests diminishing returns at higher stage

counts and highlights the importance of avoiding oversized pump designs that increase power consumption without proportional production benefits.

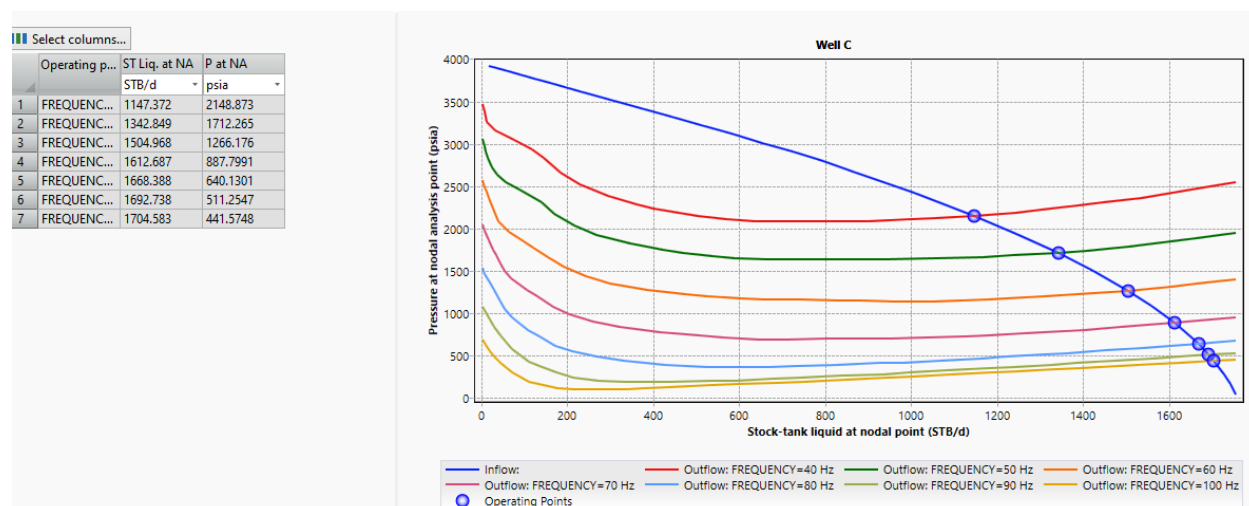


Figure 36. Nodal Analysis of Well C under ESP Frequency Sensitivity (Year 1)

Figure 36 illustrates the effect of ESP operating frequency on the nodal operating point of Well C. Increasing frequency results in higher pump head and increased production rates; however, operation at elevated frequencies (above standard design limits) significantly increases mechanical stress, heat generation, and the risk of motor failure. While higher frequencies may marginally improve production, they do so at the expense of long-term reliability and operating efficiency.

Well D

To optimize artificial lift performance in Well D, a nodal analysis sensitivity study was conducted at Year 2 operating conditions, corresponding to the onset of artificial lift installation. Given the well's low oil production rate, vertical geometry, and relatively low gas-oil ratio, a sucker rod pump (SRP) was selected as the most suitable artificial lift method. The sensitivity analysis focused on evaluating the impact of pump speed (strokes per minute) on the well operating point and oil production performance, while keeping key pump parameters, stroke

length, plunger diameter, and rod diameter, constant. This approach allows identification of an operating condition that maximizes production while avoiding pump-off conditions and excessive mechanical loading.

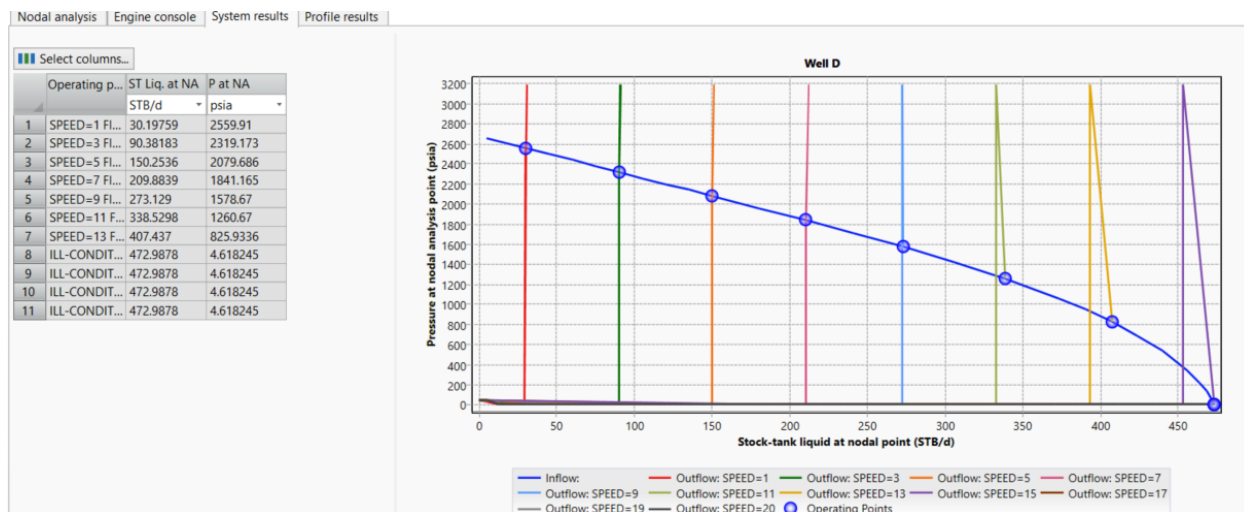


Figure 37. Nodal Analysis of Well D under Sucker Rod Pump Speed Sensitivity (Year 2)

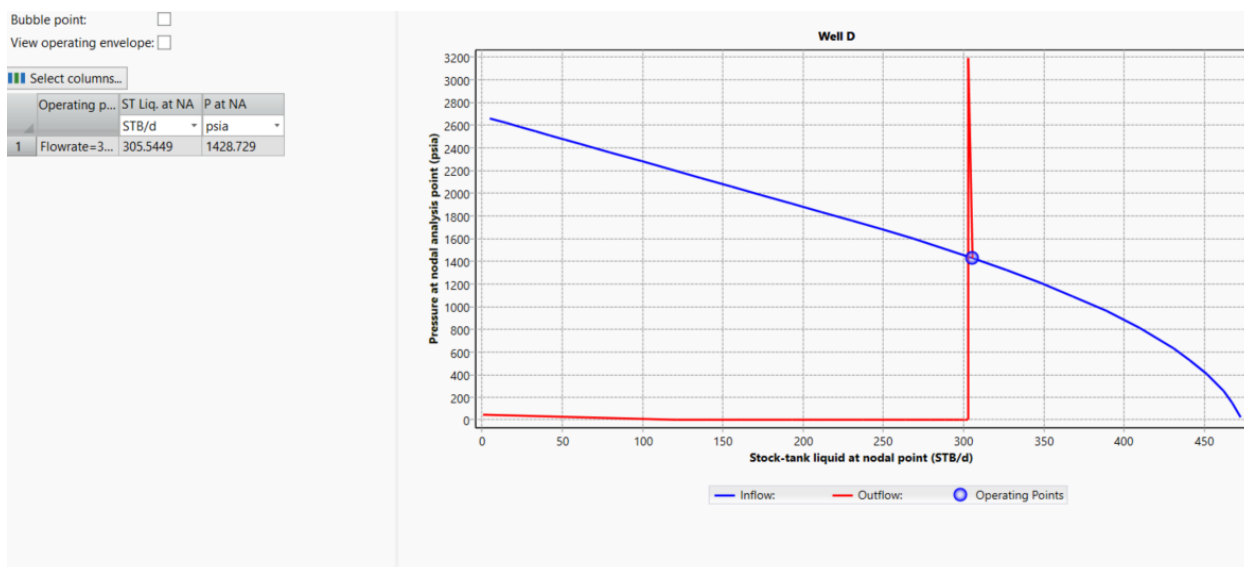


Figure 38. Optimized Nodal Operating Point for Well D at 10 Strokes per Minute (Year 2)

For Well D, a nodal analysis sensitivity study was performed at Year 2 to optimize sucker rod pump operating speed. Increasing pump speed increases liquid production by enhancing

volumetric displacement; however, excessive speeds resulted in ill-conditioned solutions, indicating pump-off and insufficient reservoir inflow. An operating speed of 10 strokes per minute was selected as the optimal configuration, providing a stable operating point with improved oil production while maintaining mechanical reliability and avoiding pump-off conditions.

Well E

To optimize the artificial lift design for Well E, a nodal analysis sensitivity study was conducted at the year corresponding to artificial lift installation. Given the well's high liquid rate and increasing water cut, an electric submersible pump (ESP) was selected and evaluated by varying both the number of pump stages and operating frequency. This sensitivity analysis was performed to identify an operating configuration that maximizes oil production while maintaining stable operating conditions and avoiding excessive drawdown or inefficient pump operation.

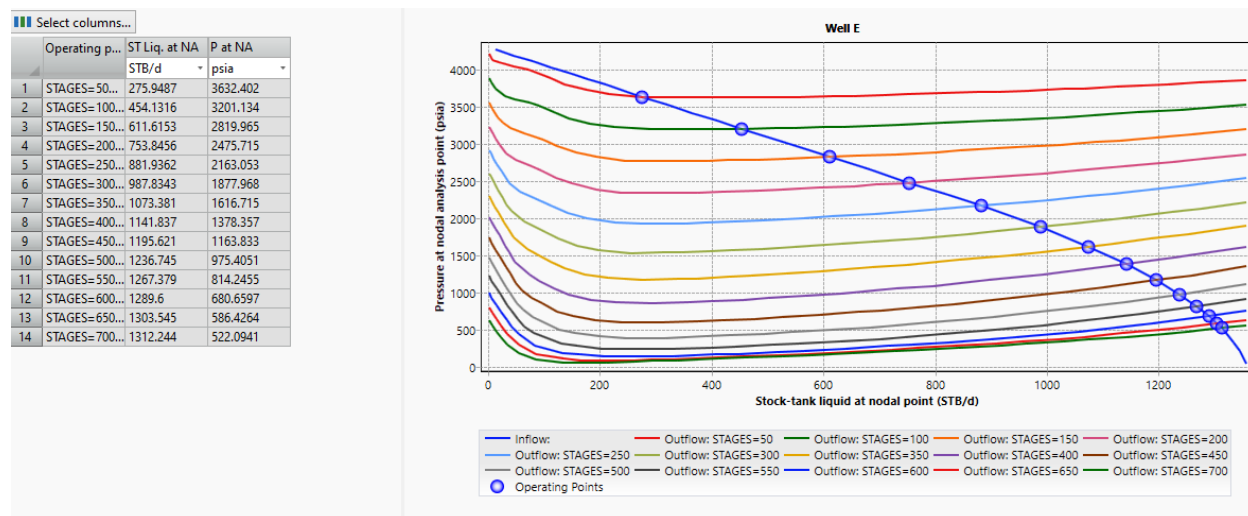


Figure 39. ESP Stage Sensitivity Analysis for Well E

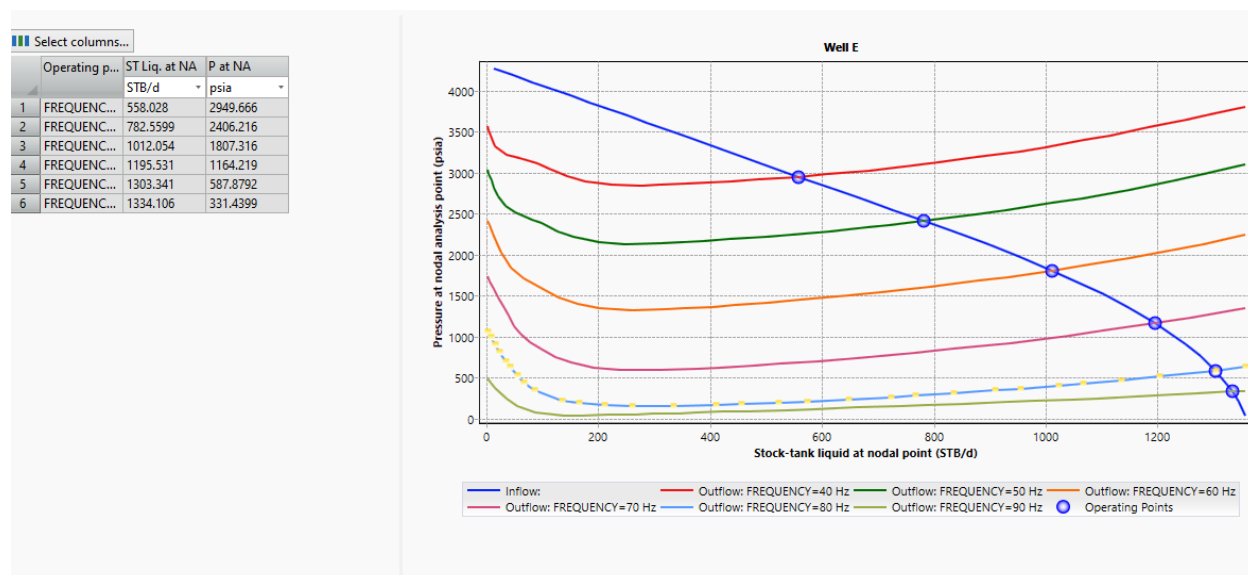


Figure 40. Optimized ESP Operating Point for Well E at 60 Hz and 450 Stages

Figure 39 shows that increasing the number of ESP stages shifts the outflow performance upward, resulting in higher achievable liquid production rates; however, the incremental production gains diminish at higher stage counts. Beyond approximately 450 stages, additional stages provide limited improvement while increasing system complexity and power demand. Figure 40 highlights the selected operating point at 60 Hz and 450 stages, which delivers a favorable balance between production performance, pump efficiency, and operational reliability. This configuration avoids excessive operating frequency while achieving near-maximum oil production, making it the optimal ESP design for Well E.

Well F

To determine the optimum sucker rod pump operating conditions for Well F, a nodal analysis sensitivity study was conducted at Year 1 operating conditions, corresponding to the onset of artificial lift installation. Pump speed was varied while keeping stroke length, plunger diameter, and rod diameter constant to evaluate the impact of strokes per minute (SPM) on the well operating point and oil production performance.

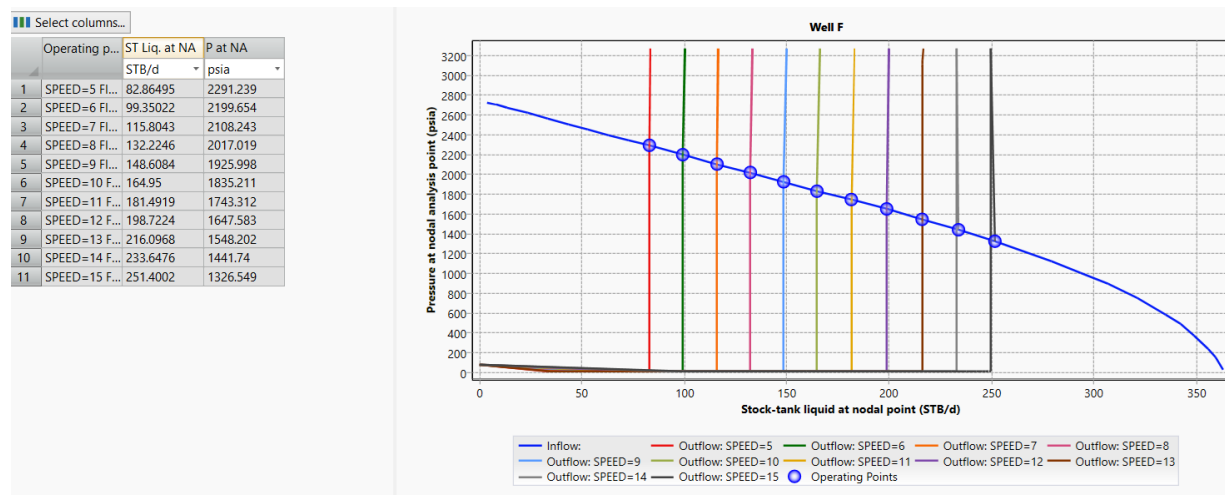


Figure 41. Nodal Analysis of Well F under Rod Pump Speed Sensitivity (Year 1)

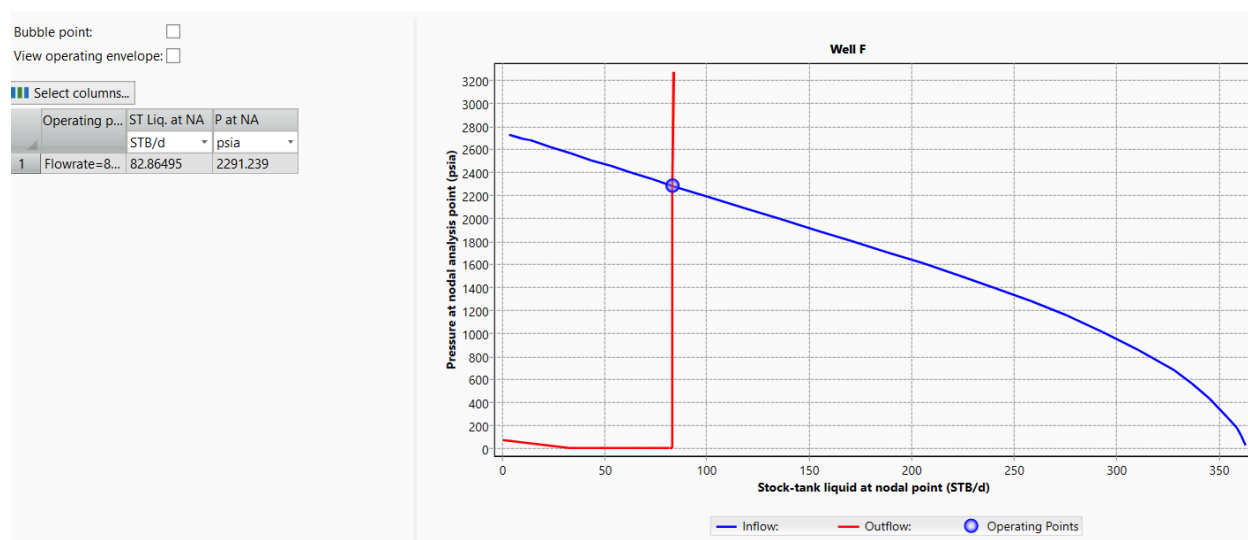


Figure 42. Optimized Nodal Operating Point for Well F at 5 Strokes per Minute (Year 1)

Figure 41 shows that increasing pump speed shifts the outflow curve toward higher liquid production rates; however, the incremental production gains diminish rapidly at higher speeds. At elevated strokes per minute, the operating point approaches conditions associated with excessive drawdown and increased risk of pump-off, which is undesirable for a low-rate well such as Well F. Figure 42 highlights the selected operating point at 5 strokes per minute, which delivers approximately 83 STB/d while maintaining a relatively high flowing bottomhole

pressure. This operating condition provides a stable and conservative production rate, minimizes mechanical stress on the pumping system, and reduces the likelihood of gas interference or premature equipment failure. Therefore, 5 spm was selected as the optimum rod pump speed for Well F.

Uncertainty Associated with Multiphase Flow Modeling

Multiphase flow modeling plays a critical role in predicting pressure losses, flow regimes, and velocity profiles in production tubing and pipelines. However, different multiphase flow correlations are developed using varying experimental datasets, assumptions, and flow regime maps, which introduce uncertainty into hydraulic predictions (Ali et al., 2024; Daraboina et al., 2018). This section evaluates the uncertainty associated with multiphase flow model selection and examines whether the choice of correlation influences tubing and pipeline sizing decisions. Well B is used as a representative example.

Effect of Multiphase Flow Model Selection

In this study, pressure and superficial gas velocity profiles for Well B were generated using PIPESIM's steady-state multiphase flow framework. It is well established in the literature that different correlations, such as Beggs & Brill, Hagedorn & Brown, and Mukherjee & Brill, can yield different predictions of pressure losses and flow behavior under identical operating conditions due to their underlying assumptions regarding slip velocity, flow regime transitions, and frictional losses (Beggs & Brill, 1973; Hagedorn & Brown, 1965; Mukherjee & Brill, 1985).

Previous studies have shown that discrepancies between multiphase flow models tend to become more pronounced at high gas-liquid ratios and in low-pressure regions near the surface, where gas expansion dominates flow behavior (Shoham, 2006; Daraboina et al., 2018). In the context of this project, Well B exhibits increasing superficial gas velocity toward the surface,

indicating operating conditions where such model sensitivity may be expected. However, despite potential differences in absolute pressure predictions reported in the literature, the qualitative trends in pressure decline and gas expansion behavior are generally consistent across models.

Impact on Tubing and Pipeline Sizing Decisions

Although absolute pressure predictions differ between multiphase flow models, the relative ranking of tubing and pipeline diameters remains unchanged. In other words, while one model may predict slightly higher or lower pressure losses, all models consistently identify the same diameter ranges as hydraulically favorable. This observation aligns with findings in the literature, which suggest that multiphase flow model uncertainty typically affects absolute values rather than qualitative design decisions (Brill & Mukherjee, 1999; Shoham, 2006).

Therefore, while multiphase flow uncertainty introduces variability in predicted pressure and velocity profiles, it does not significantly alter the final tubing or pipeline sizing decisions for Well B in this study. Conservative design margins are sufficient to accommodate these uncertainties.

Discussion of Model Uncertainty

The uncertainty associated with multiphase flow modeling stems from three primary sources: limited experimental data at field-scale conditions, empirical flow regime maps, and simplifying assumptions regarding phase interaction. As noted by Shoham (2006), no single multiphase flow correlation can accurately predict all flow conditions, and model selection should be guided by well geometry and operating conditions.

Recent studies emphasize that using multiple correlations and comparing results is considered best practice in production system design, particularly for wells with high gas–liquid

ratios or deviated geometries (Ansari et al., 2014). In this context, awareness of model-related uncertainty provides confidence that the selected tubing and pipeline sizes are robust against reasonable variations in predicted pressure losses.

Overall, while multiphase flow model selection introduces uncertainty in predicted pressure and velocity profiles, this uncertainty is not expected to significantly alter the tubing and pipeline sizing decisions made in this study. Conservative design margins and consistency with established design practices help mitigate the impact of model-related uncertainty on production system performance.

Economic Evaluation

Economic Assumptions

Parameter	Value	Unit
Oil Price	50	\$/bbl
Water Disposal	0.5	\$/bbl
ESP Cost	0.75	\$/bbl (Total Liq)
Rod Pump Cost	0.3	\$/bbl (Total Liq)
Gas Lift Cost	1.5	\$/Mscf
Days in Year	365	Days

Figure 43. Economic assumptions used in the project

A simplified economic model was used to evaluate the profitability of each well. Oil revenue was calculated using a constant oil price of \$50/bbl, while operating expenditures include water handling and artificial lift energy costs where applicable. Capital expenditures were not explicitly modeled, and the analysis focuses on operating profitability trends.

Annual Cash Flow Analysis

Well A							
Parameter	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Gross Revenue (\$)	10566416.03	7860424.65	7860424.65	6562244	6915943	6042809	5261174
Water Cost (\$)	14408.74913	15452.69219	16420.22944	17167.27	22909.76	25725.25	29378.63
Energy Cost (\$)	0	0	0	0	1368750	1368750	1368750
Total OPEX (\$)	14408.74913	15452.69219	16420.22944	17167.27	1391660	1394475	1398129
Net Profit (\$)	10552007.28	7844971.958	7844004.421	6545076	5524284	4648333	3863045
Well B							
Parameter	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Gross Revenue (\$)	20080694	13984297.93	11700337.8	8347074	5715469	4082593	3093886
Water Cost (\$)	35436.51882	30697.23935	32235.62455	29205.74	25803.23	24312.96	25099.4
Energy Cost (\$)	0	0	0	0	547500	547500	547500
Total OPEX (\$)	35436.51882	30697.23935	32235.62455	29205.74	573303.2	571813	572599.4
Net Profit (\$)	20045257.28	13953600.69	11668102.18	8317868	5142166	3510780	2521287
Well C							
Parameter	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Gross Revenue (\$)	31413414.75	29434859.25	17616971.38	11276702	6062535	3727272	2645522
Water Cost (\$)	169149.1563	213148.9808	179011.1607	172574.6	160441.4	251466.1	#DIV/0!
Energy Cost (\$)	0	761246.3599	532771.3117	428012.5	331600.2	433108.3	#DIV/0!
Total OPEX (\$)	169149.1563	974395.3407	711782.4725	600587.1	492041.6	684574.5	#DIV/0!
Net Profit (\$)	31244265.59	28460463.91	16905188.9	10676115	5570493	3042698	#DIV/0!
Well D							
Parameter	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Gross Revenue (\$)	4772906.075	3210545.475	5576194.425	5692989	5804573	5906839	5623256
Water Cost (\$)	18561.3014	16246.13373	37672.94893	53365.15	80358.75	135703	286806.9
Energy Cost (\$)	0	0	56060.93591	66177.03	83042.69	116862.8	83042.69
Total OPEX (\$)	18561.3014	16246.13373	93733.88484	119542.2	163401.4	252565.8	369849.5
Net Profit (\$)	4754344.774	3194299.341	5482460.54	5573447	5641172	5654273	5253406
Well E							
Parameter	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Gross Revenue (\$)	12493192.63	18470971	13703963.33	9575018	6222608	4327756	0
Water Cost (\$)	102217.0306	216833.1378	252277.5067	334781.2	868187.7	#DIV/0!	0
Energy Cost (\$)	0	602314.2717	583975.7099	645797.1	1395621	#DIV/0!	0
Total OPEX (\$)	102217.0306	819147.4096	836253.2165	980578.2	2263808	#DIV/0!	0
Net Profit (\$)	12390975.59	17651823.59	12867710.11	8594439	3958799	#DIV/0!	0
Well F							
Parameter	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Gross Revenue (\$)	1210085.23	1512285.338	1534419.485	1566067	1577553	1599172	1621298
Water Cost (\$)	5186.079557	8506.605023	11670.2327	16749.7	25967.51	47090.99	139374.7
Energy Cost (\$)	0	14177.67504	16208.65653	19386.22	25045.82	37849.62	93352.59
Total OPEX (\$)	5186.079557	22684.28006	27878.88923	36135.92	51013.34	84940.61	232727.3
Net Profit (\$)	1204899.15	1489601.057	1506540.596	1519932	1526540	1514231	1388571

Figure 44. Cash flow calculations and results

The annual cash flow results indicate that all wells are economically viable during the early production years, driven by high initial oil production rates and minimal operating costs under natural flow conditions. Energy costs are zero for years without artificial lift installation, as no external power or compression is required to sustain production. Once artificial lift is installed, energy costs increase due to power consumption associated with gas lift, ESPs, or rod pumps, leading to higher total OPEX. Despite the added energy expenses, artificial lift extends productive well life and stabilizes oil rates, resulting in sustained net profit in later years. Overall, the project remains economically feasible, with artificial lift improving cumulative recovery and long-term profitability.

Conclusion

An integrated production system assessment was carried out for six wells, combining nodal analysis, multiphase flow simulation, artificial lift optimization, and economic analysis. Under natural flow, the tubing sizing study showed that enlarging tubing diameter reduces frictional pressure losses only up to a certain threshold, beyond which hydrostatic head becomes dominant and additional diameter provides limited production benefit. Consequently, each well was assigned an optimum tubing size rather than an oversized completion, in line with nodal analysis principles (Economides et al., 2013).

Flow stability evaluation underscored the need to maintain sufficient mixture velocity to avoid liquid loading. Wells E and F showed low-velocity, unstable flow behavior in natural depletion, supporting the need for early artificial lift. Rod pumps and ESPs were selected based on fluid rate, depth, and long-term reliability, consistent with standard artificial lift selection practices (Takács, 2015).

Gas lift sensitivity studies for Wells A and B indicated that higher gas injection rates increase production only up to an optimum, after which further gas injection produces negligible incremental oil. This reinforces that gas lift design must trade off added production against compression and operating costs (Brown, 1984). For the pumped wells, variations in pump speed, frequency, and number of stages were analyzed to secure stable operation while preventing excessive reservoir drawdown.

The economic analysis showed that energy costs are effectively zero during natural flow and rise only after artificial lift is installed, explaining the absence of energy expenditures in early years. Even with higher operating costs, artificial lift extended field life and maintained positive net cash flow for all wells. Overall, the findings demonstrate that integrated nodal

analysis, disciplined timing of artificial lift, and conservative flow assurance criteria yield technically robust and economically resilient production system designs under multiphase flow conditions (Shoham, 2006; Ansari et al., 2014).

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